

# When do regulatory interventions work?

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## Abstract

We study two instances when an orders-to-trades ratio fee was used as an intervention to slow down algorithmic trading in the Indian equity market. In the first instance, the exchange used this fee to manage high order placement load on limited bandwidth. In the second instance, the regulator used this fee to address public policy concerns that a high orders-to-trades ratio harms market quality. We develop difference-in-difference estimations that exploit micro-structure features in order to identify the causal impact of the fee in each case. We find a lower aggregate market orders-to-trades ratio, higher market liquidity with lower liquidity volatility when the exchange used the fee. But there was little or no change in the orders-to-trades ratio or market quality when the regulator used the fee. We conclude that when such interventions are used to solve a stated market failure, it is more likely to deliver better market outcomes.

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# 1 Introduction

High levels of trading activity in financial markets has been frequently viewed with scepticism from policy makers. Such concerns became exacerbated since the global credit crisis of 2008, and with the growing dominance of high-frequency trading. Often the concerns have lead to policy interventions to curb the trading activity. The policy interventions have taken different forms, from technical barriers that slow down the rate of order placement into the trading systems, to charges on order placement. An early example of an intervention against high trading activity is a tax on transactions (Tobin, 1978). A more recent example is the *orders-to-trades ratio* (OTR) fee. This fee is charged to a market participant when her ratio of orders to trades crosses a fixed threshold.

Over the previous decade, several exchanges have experimented with the use of an OTR fee to control high frequency trading starting with the Chicago Mercantile Exchange which implemented a fee in 2005. Recent literature documents the impact of these fees at exchanges in Canada, Italy and Norway (Friederich and Payne, 2015; Jorgensen *et al.*, 2017; Capelle-Blancard, 2017). Such interventions are often found to have adverse effects on the market. For example, when the Scandinavian countries imposed a transactions tax on equity trading in the 1980's, local trading activity and price discovery migrated to competing financial markets in the Euro-zone. The magnitude of the impact of the OTR fee varies across exchanges. However, there is more evidence of deterioration in market liquidity than there is evidence of improvement in market liquidity or a decrease in market volatility, after such a fee is put in place.

In this paper, we argue that when the intervention is designed to rectify a clearly stated market failure, it is more effective in achieving desired market outcomes. We study two unique instances in the Indian equity markets to analyse the causal impact of the OTR fee that was applied to curb algorithmic trading. In the first instance, an exchange used the OTR fee because bandwidth was sparse and early adopters of algorithmic trading were able to dominate the market while placing orders into the limit order book. In the second instance, the securities market regulator used the OTR fee in response to public policy concerns about algorithmic trading. In both cases, the fee was targeted to reduce the amount of algorithmic trading in the market.

We use the event study methodology to understand how an OTR fee affects the average OTR in the market. We innovate a strategy to isolate the causal impact of the fee by exploiting some unique micro-structure features of the Indian equity markets. First, we use the fact that the OTR fee was only applied to derivatives and not on the underlying spot markets. This suggests that the underlying spot can be used as the *control* with the derivatives being the *treated* set in the event analysis of the impact of the OTR fee.

But there is a possible endogeneity bias in using derivatives and spot as treated and control. Derivatives and the underlying are exposures on the same asset but with different trade-offs on leverage and liquidity. When the costs of trading increases on one, trading will shift to

the other (Brunnermeier and Pedersen, 2009; Aggarwal and Thomas, 2011). An OTR fee on the derivative can have an indirect effect on the market quality of the underlying spot. In order to adjust for this endogeneity bias, we use a second micro-structure feature in the design of our analysis: not all stocks have derivatives trading on them. If stocks with derivatives become more liquid after the OTR fee is implemented, then these derivative-linked stocks will become cheaper to trade than stocks without derivatives.

We create a *second* treated set of the underlying stocks on which derivatives are traded, where the control are *matched* stocks which do not have derivatives trading on them. If the OTR fee has a causal impact on the derivatives relative to the underlying spot *and* the OTR fee also has an impact on the stocks with derivatives relative to a matched stock without futures, this helps to support the claim of the causal impact of the OTR fee.

We use these two sets of treated-control securities in separate difference-in-difference estimations in order to establish the casual impact of the OTR fee. The OTR fee was implemented once when the exchange imposed the fee in 2009, and again when the market regulator imposed the fee in 2012. In both cases, the fee was imposed only on derivatives. We study the single stock futures because they are more liquid compared to single stock options in the Indian markets. The variables of interest include the OTR level, market liquidity measured by the estimated impact cost of transactions for two different sizes, and market efficiency measured by the variance ratio of returns and the volatility of the impact cost of the transaction.

We find that the first fee intervention in 2009 was effective. The aggregate market OTR level in the futures *decreased* relative to the OTR in the underlying spot after the exchange started imposing the fee, while OTR of the underlying spot *increased* relative to matched control stocks. This strongly suggests that the 2009 OTR fee caused the OTR to drop. Market liquidity of the treated stocks *improved* after the fee was imposed in 2009. The spread and impact cost *decreased*, as did the volatility of liquidity. The results suggest that the OTR fee improved market quality, when it was applied to manage the use of limited exchange bandwidth in 2009. In contrast, related studies on the impact of the OTR fee only report a negative or no impact, on market liquidity.

We do not find a clear causal impact for when the fee was implemented in 2012. There appears to be an *increase* in the OTR level of the single stock futures relative to their underlying as control and *decrease* in the OTR level of the underlying relative to the stocks without futures as the control. This is counter to the regulatory objective of reducing the OTR in the market. One reason for this could be that the fee in 2012 was only applicable to orders placed beyond a one percent price band from the best bid-ask price. An analysis of the limit order book away from the touch does suggest that there is a significant decrease in the OTR among the higher prices in the book. Among the market quality measures, the only significant evidence is that the volatility of liquidity *increased* after the fee, particularly for orders that have prices away from the bid-ask price. Recent studies show that investors demand a higher premium for lower volatility of liquidity rather than just lower liquidity (M.Blau *et al.*, 2018).

We infer from these results that the 2009 fee was more visible and effective, while the 2012 fee had ambiguous results. The 2009 fee was effective in reducing the dominance of algorithmic trading (lower OTR) while simultaneously improving market quality (lower transactions costs, lower volatility of transaction costs). In comparison, the 2012 fee did not have any clear impact on the dominance of algorithmic trading, even though there was some impact on the market quality (higher volatility of transaction costs at larger trade sizes). As a regulatory intervention on algorithmic trading, the fee only delivered visibly positive outcomes in the 2009 implementation.

We propose that the difference in the effectiveness of these two implementations is driven by the clarity of the stated objectives for the interventions. The stated objective of the fee in 2009 was to manage exchange access bandwidth. This was a well-designed fee with a specific target, and the target was achieved. The fee of 2012 was motivated by public policy concerns, which did not lead to a well specified design of the fee. Our results support the position that clearly stated regulatory objectives are likely to achieve desired outcomes. Jorgensen *et al.* (2017) similarly suggest a link between the design of the intervention and success in achieving a desirable regulatory outcome.

The paper is organised as follows: Section 2 provides the context of how high frequency trading has attracted regulatory interventions despite mounting evidence that market liquidity improves with increased levels of high frequency trading. Section 2.2 discusses existing research on applications of the OTR fee as a regulatory intervention to manage the effects of high frequency trading, specifying instances when it has been used in the Indian equity markets and the research questions we ask in the paper. Section 3 describes the methodology and data used to measure the causal impact of the OTR fee. Section 5 describes the results of the analysis, and Section 6 concludes with what we learn along with some suggestions on future research in this area.

## 2 High frequency trading and regulatory interventions

Algorithmic trading (AT) and high frequency trading (HFT) have been the dominant method of trading in limit order book exchanges ever since early 2000. In a limit order book exchange, there is transparency about the prices and quantities that are available at which to buy and sell any given stock. Thus, limit orders are like free options to the overall market (Harris and Panchapagesan, 2005). But, traders who offer these limit orders face the adverse selection risk that their orders may be picked off by better informed traders or opportunistically by other traders.

Traditionally, limit order traders have used a variety of strategies to reduce their exposure to such risk, which includes hiding their true order size or posting prices away from the market. Advances in technology have allowed these traders to protect their orders by easily

modifying and canceling them using algorithms that respond when information arrives. This leads to lower adverse selection risk, which allows limit order traders to compete more on price as well as on size. This, in turn, leads to improved available liquidity in limit order book exchanges.

The additional benefit of technology in these exchanges is in allowing a greater degree of transparency about order flow and trades, and enabling researchers to measure and document the quality of market outcomes on these exchanges. The resulting research increasingly shows that the dominance of AT has improved market quality on average.

Earlier, the focus of most of the research was on U.S. exchanges, where evidence of the impact of AT and HFT was mostly positive (Angel *et al.*, 2011; Robert *et al.*, 2012; Avramovic, 2012; Easley *et al.*, 2012; Cumming *et al.*, 2012; Weisberger and Rosa, 2013; Bollen and Whaley, 2014). For example, Hasbrouck and Saar (2013) studied the effect of low latency AT over two distinct periods at the NASDAQ and found that higher levels of low latency activity correlated with better market quality. Hendershott *et al.* (2011) found that the NYSE auto-quoting facility, that was introduced in 2003, reduced effective spreads for all stocks, particularly for large cap stocks.

Since U.S. markets tend to be complex, fragmented networks of both transparent, open, nation-wide exchanges and local, dark pools and electronic crossing networks, there was initially some scepticism that this research could not be globally applicable to financial markets where the micro-structure is different. However, there is now a growing body of research on non-U.S. exchanges as well. For example, both the Deutsche Borse in Germany and the National Stock Exchange in India publish orders and trades data that are explicitly tagged as AT. Hendershott and Riordan (2009) studied the data of the Deutsche Borse Xetra platform and found that AT contributes to discovery of prices and does not contribute to excess volatility. Others found that trading latency was lower, liquidity was higher and adverse selection was lower once the trading system at the Deutsche Borse was upgraded (Hendershott and Moulton, 2011; Hendershott and Riordan, 2013). Aggarwal and Thomas (2014) studied small and medium market cap stocks at the National Stock Exchange of India (NSE) that had a higher fraction of AT compared to control stocks, and found that stocks with higher AT intensity had higher liquidity, lower intra-day volatility of liquidity, lower volatility, and a lower likelihood of flash crashes compared to similar stocks with a lower fraction of AT. Thus, even the more recent evidence from other limit order book exchanges in the world indicate that a higher fraction of AT improves market quality.

Despite all this evidence, there has remained substantial public discomfort and regulatory concerns about the effects of AT and HFT as these have increasingly dominated the trading on limit order book exchanges.

Inevitably, poorly constructed algorithms and ill-tested systems have occasionally been found to bring exchange trading to a halt in the middle of a trading day. These have been viewed as systemic problems rather than episodic signs of market participants adjusting to AT. Some of these, with extremely temporary extreme price movements, and others,

which have led to market closure during trading hours, have become part of the everyday parlance. The better known examples include the 6<sup>th</sup> May 2010 ‘Flash Crash’ in the U.S. markets, the crash at Tokyo Stock Exchange triggered by excessive trading of *Livedoor* stock (Brook, 2005), and the crash at the NSE because of a fat-finger trade in the “Nifty” index futures (Aggarwal, 2017). Even exchanges and securities firms, which continuously invest in capacity to handle new pressures on their systems because they benefit from the higher turnover that HFT brings, have to trade off higher revenues against the cost of technical errors and market closures when their trading systems are overwhelmed by HFT messaging traffic and the occasional trader error.

These episodes have led to a persistent, ubiquitous discomfort of extreme and market-wide consequences of errors in using AT and HFT, which finds voice in the form of *public policy concerns* about HFT. Thus, despite research that HFTs are not the cause of such crashes (Kirilenko *et al.*, 2014), regulators and exchanges continue to propose interventions that act as disincentives to HF traders for the potential negative externalities that they can impose on markets.

Because the objective of such interventions is to reduce order placement activity in the exchange, it is likely to have a negative effect on liquidity and, in turn, a negative effect on the informational efficiency of price. However, these interventions are designed to improve long-term investor confidence by reducing the chances of HFT being the source of an unexpected trading closure on exchanges, such as the Flash Crash of 2010. If the intervention is effective, it will raise investor confidence, which over the longer horizon, should lead to greater trading, as compared to if these interventions had not been implemented.

## 2.1 A fee on high orders-to-trade ratios (OTR)

Of the many suggested interventions, the two that are most widely implemented are those that seek to slow down HFT using barriers in the trading mechanism, and those that impose a penalty or fee on using HFT. An example of the first is a minimum resting time for orders before any action can be taken on them. Another is the introduction of a random delay between order arrival and order processing, which introduces uncertainty in the latency of order placement and is likely to prevent a monopoly outcome among trading firms that chase cutting edge hardware systems in order to reach lowest latency Harris (2013). In 2016, the U.S. Securities Exchange Commission (SEC) granted an exchange license to the IEX,<sup>1</sup> which explicitly introduces a delay of 350 micro-seconds (called a ‘speed bump’) on orders coming to the exchange, to ensure that prices available for trade are not stale.

The other common intervention is to charge HFT for canceling or modifying orders within a short period, so that no other trader can act on the HFT order. One such intervention is the orders-to-trade ratio (OTR) fee. This is a charge or fee for all order placement or trade execution strategies that generate a high orders to trade ratio for a given order placed.

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<sup>1</sup><https://iextrading.com/>

If the fee achieves its objective, then it will disincentivise frivolous or mischievous order placement, which may lead to lower liquidity (expected cost) but higher informational efficiency of prices (expected benefit). Such a fee was first put in place by the Chicago Mercantile Exchange in April 2005, where the fee was charged if the OTR exceeded a threshold of 25 : 1.

The empirical literature on the impact of the OTR fee, that have been implemented by various exchanges around the world, suggests that the costs outweigh benefits. For example, Friederich and Payne (2015) and Capelle-Blancard (2017) both found that a OTR fee imposed by the Italian Stock Exchange led to decreased trading activity in the aggregate. Jorgensen *et al.* (2017) examined the Oslo stock exchange data and found the fee did not cause any adverse changes to average market liquidity.

We capitalise on a unique opportunity to observe two instances when an OTR fee was imposed in the Indian equity derivatives market, but with different objectives. In the first instance, the fee was imposed by the exchange to better manage the high messaging load on their trading systems. In the second instance, the fee was imposed by the regulator as a response to broad public policy pressure on regulating HFT. In both cases, the structure of the intervention was applied on the same market, but there were differences in how the fee was imposed, and for whom it was relevant. We can thereby identify how a regulatory objective can effect the outcome of the intervention.

## 2.2 OTR fee regimes in the Indian equity markets

The exchange studied are the equity spot and derivatives markets at the National Stock Exchange of India (NSE). The NSE is one of two equity stock exchanges in India,<sup>2</sup> with a market share of 75% of the equity spot market, and about 98% of the equity derivatives market (SEBI, 2013). Trading takes places from 9am to 3:30 pm through an anonymous continuous electronic limit order book mechanism, where orders are matched on a price time priority.<sup>3</sup> More than 1400 securities are listed at the equity platform of NSE, out of which 146 securities are traded on the derivatives platform. The selection of securities on the derivatives platform is based on the free float market capitalisation, average traded value and the price impact of a trade. The Securities and Exchanges Board of India (SEBI) is the securities markets regulator, which tends to be prescriptive in setting rules and regulations. The exchanges have the same trading times and follow the same parameters in introducing products on the exchange, as well as in how trading, clearing and settlement is implemented.

AT was first permitted by SEBI in equities in April 2008. Soon after, exchange members started implementing and experimenting with the use of AT. The NSE put a fee on high

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<sup>2</sup>The other stock exchange is the Bombay Stock Exchange, BSE.

<sup>3</sup>The opening price is determined through a pre-open call auction mechanism conducted between 9am to 9:15am.



OTR in equity derivatives in 2009 in order to reduce the load on exchange infrastructure. The exchange detected that there was a very high rate of Immediate and Cancel (IOC) orders that were used by traders to execute arbitrage strategies on derivative markets. A large proportion of these orders remained unexecuted but added significant load on the bandwidth.

In imposing a fee on orders-to-trades that were higher than a given threshold, the exchange acted in their capacity as a self-regulatory organisation that was seeking to ensure good market quality to stakeholders in the equity markets. The circular issued by the exchange stated the objectives of the fee as follows:

“Of late, it is observed that the Order to Trade ratio in the F&O segment has been increasing significantly. Based on the analysis of the same, it has been observed that some trading members have been placing very large number of unproductive orders which rarely result into trades in the F&O segment which leads to increase in latency in order placement and execution for the other members. Such members are observed to have very large order to trade ratio which is significantly higher than the market average. In order to prevent such system abuse and to ensure fair usage of the system by all the members, it has been decided to levy a charge to deter system abuse in the F&O segment with effect from 1st October, 2009 as per the slabs below.”

The fee was applicable only on equity derivatives. It was implemented uniformly across all market participants and all order types, without any exceptions. In July 2010, based on a ‘notable’ improvement in the OTR in the derivatives segment, the exchange reduced the fee and raised the minimum thresholds for daily OTR.

Even though AT was permitted in 2008, the intensity of AT increased significantly as a percentage of the total trading only after the exchanges implemented *co-location*.<sup>4</sup> When AT became the larger fraction of overall market trades, SEBI decided to impose an OTR fee in 2012. Unlike the exchange intervention, the objectives for which the fee was imposed were not to manage a specified or tangible problem but were rather for a general, public policy concern. SEBI (2012) said:

“In order to ensure maintenance of orderly trading in the market, stock exchange shall put in place effective economic disincentives with regard to high daily order-to-trade ratio of algo orders of the stock broker. Further, the stock exchange shall put in place monitoring systems to identify and initiate measures to impede any possible instances of order flooding by algos.”

When SEBI implemented the fee in July 2012, it was levied in lieu of the *increased* use of algorithms by market participants. Thus, the fee was applicable *only* on algorithmic orders, and there were several exceptions within that. For example, all order entries, that

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<sup>4</sup>Aggarwal and Thomas (2014) shows how AT intensity started from around 20 percent of the market in 2010 to 55-60 percent by 2013, with some stocks having a AT intensity of 70 percent.

were placed or modified within one percent of the last traded price, were exempt from the fee. Orders from trading members who were designated as market makers were exempt from the fee.<sup>5</sup> The stated explanation for the exemptions was that the regulator wanted to minimise any adverse impact of the fee on the available liquidity at the best bid and ask prices in the limit order book. There was a further modification of fees in May 2013, when SEBI directed exchanges to double the magnitude of the fee SEBI (2013).<sup>6</sup>

**Table 1** Details of the instances of OTR fee implementation

| 2009-10                                                                                                                                                | 2012-13                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• By the exchange on equity derivatives</li> <li>• on all participants</li> <li>• on all order types</li> </ul> | <ul style="list-style-type: none"> <li>• By the regulator on equity derivatives</li> <li>• <i>not applicable</i> to participants who are market makers</li> <li>• <i>only</i> on algo orders</li> <li>• <i>only</i> on orders <i>outside</i> <math>\pm 1\%</math> LTP</li> <li>• with an additional penalty of a trading ban on the first 15 minutes on the next trading day if (OTR &gt; 500)</li> </ul> |

The details of the OTR fee implementation in both regimes are summarised in Table 1. Figure 1 presents a graph of the timelines involved. For a frame of reference, the graph also shows the rise of AT in single stock futures (SSF). Unlike in the U.S. equity markets,

<sup>5</sup>In India, designated market makers are only for the illiquid indices. The stocks covered in this study did not have any designated market maker under the *Liquidity Enhancement Scheme* (LES) under which exchanges were permitted to pay trading members a fee for maintaining two-way bids on select derivative contracts.

<sup>6</sup>The SEBI circular in 2013 is more specific about the nature of the disincentive compared to the circular in 2012. The 2013 circular states:

“4. As directed vide circular dated March 30, 2012 stock exchanges have implemented a framework of economic disincentives for high daily order-to- trade ratio of orders placed from trading algorithms by prescribing penalties in form of ‘charges to be levied per algo orders’ at various levels of daily order-to- trade ratio. The penalty rates specified by the stock exchanges have been reviewed and in order to provide sufficient deterrence, stock exchanges are directed to double the existing rates of ‘charges to be levied per algo orders’ specified in their circulars / notices.

5. In order to discourage repetitive instances of high daily order-to-trade ratio, stock exchanges shall impose an additional penalty in form of suspension of proprietary trading right of the stock broker / trading member for the first trading hour on the next trading day in case a stock broker / trading member is penalized for maintaining high daily order-to-trade ratio, provided penalty was imposed on the stock broker / trading member on more than ten occasions in the previous thirty trading days.

6. The circular shall be applicable with effect from May 27, 2013.”

single stock derivatives trading is concentrated on single stock futures and not on single stock options. Thus, we focus our analysis only on the SSF.

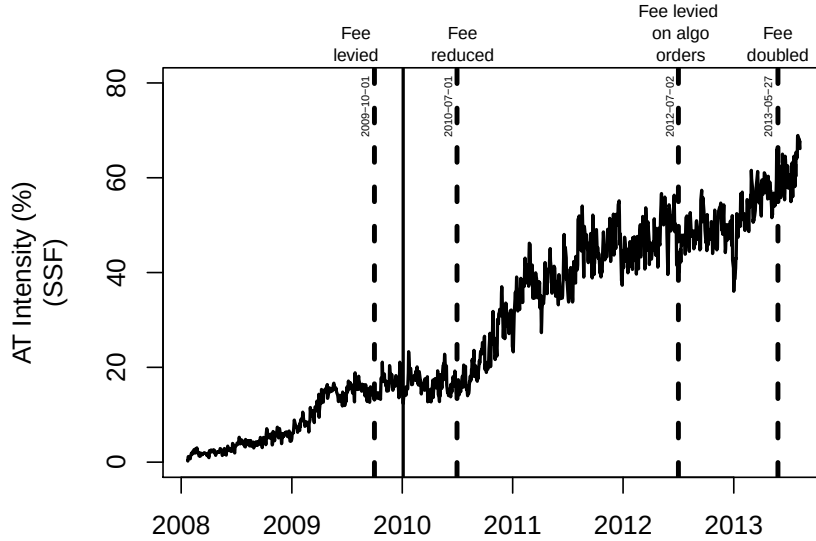
In the graph, the solid vertical line represents the date on which co-location services commenced (January 2010). The dashed horizontal lines mark the various dates of an OTR-fee based interventions. The first line denotes when the exchange first imposed the OTR fee, while the second line denotes when the exchange reduced the fee. The third line denotes when SEBI re-introduced the fee with restrictions, and the last line denotes when SEBI raised the amount of the fee. At the start of this period, the fraction of AT was relatively low at around 20 percent when the exchange imposed the fee in late 2009, compared to the period when SEBI imposed the fee, when the fraction of AT was significantly higher, at around 60 percent, for the overall market trades.

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**Figure 1** AT intensity in single stock futures at the NSE

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The graph shows the AT intensity on the single stock futures (SSF) market at NSE between 2009 and 2013. AT intensity is measured as a fraction of the total traded value of AT trades in a day relative to the total traded value on that day. The solid vertical line indicates the date on which co-location was operationalised, January 2010. The first two dotted lines indicate dates of OTR fee intervention by NSE, and the last two dotted lines indicate the dates of OTR fee intervention by SEBI.



Here, we focus on the first event when the OTR fee was imposed by NSE in 2009 (NSE, 2009), and the second event when the OTR fee was mandated by SEBI in 2012 (SEBI, 2012; NSE, 2012).

We select these two events because the design variations across these two events can help identify what makes such a fee effective. For example, while the 2009 fee was imposed on *all* orders, the 2012 fee was imposed only on algorithmic orders that were placed *outside* of the best bid and offer prices on the screen. A fee can be expected to increase the average cost of trading and to reduce the average OTR in the market. Thus, the fee that was imposed

in the 2009 period may be expected to have the effect of *reducing* the overall market-wide OTR. Typically, if an extra fee or charge reduces the number of orders into the trading system, it is likely to lead to more adverse market qualities of efficiency and liquidity. But in certain cases, excessive orders from algorithmic traders could crowd out genuine liquidity providers from the market. In such cases, a fee against traders who have an excessive order placement to trades executed ratio, that leads to better bandwidth problems by the exchange for the broader set of market participants, may lead to improvements in market efficiency as well as market liquidity on average.

On the other hand, if the fee was imposed differently for different participants or for orders in different parts of the limit order book, it would affect trading behaviour in different ways. This makes it difficult to predict how the average OTR was affected in the case when the OTR fee was re-introduced in 2012, or how it affected market efficiency and liquidity.

In the analysis, we ask the following questions:

**Q1:** Does the OTR fee have the intended impact of reducing the average level of the OTR?

**Q2:** What were the consequence of the fee on market liquidity?

**Q3:** What were the consequence in the fee on market efficiency?

In addition to presenting how the broad market responded to the OTR fee, we identify the causal impact of the fee. The methodology used for the analysis is presented in the following section.

### 3 Methodology

To estimate the causal impact of the OTR fee, we use a difference-in-differences approach applied to each of the two implementations discussed earlier. First, we locate the details of the periods of the two events. In each case, we use a three month window before and after the fee was imposed. Given the two events selected, the dates that we use in the event study analysis are:

**Event 1** (Fee imposed by NSE on October 1, 2009)

- a) Pre event period: July 2009 to September 2009
- b) Post event period: October 2009 to December 2009

**Event 2** (Fee imposed by SEBI on July 2, 2012)

- a) Pre event period: April 2012 to June 2012
- b) Post event period: July 2012 to September 2012

To exclude the possible announcement effects, we exclude the period between the announcement and the implementation of the fee from the analysis. For example, in the case of Event 1, the announcement was made on September 7, 2009 by NSE. Hence we exclude the period between September 7, 2009 to October 1, 2009 from the analysis. Similarly, we exclude the period between June 29, 2012 to July 2, 2012 from the analysis of Event 2.

For each event, we examine the behaviour of the OTR as well as a set of carefully selected market quality measures, several of which are calculated using intra-day data. For this, we use a proprietary tick-by-tick data-set of all orders and trades in the equity and the equity derivatives segment of NSE. The data include the type of order, and order details such as price and quantity, as well as categorical variables: a) trader type category (whether institutional, proprietary or neither of the two), b) if the order/trade was by an AT or non AT, and (c) the type of order event (whether it was an order entry, modification or cancellation). These data are used to create daily measures of OTR, market liquidity and market efficiency around the time of introduction of the OTR fee in both events.

### 3.1 Causal identification

Once the event windows are identified, we describe the methodology using which we identify and isolate a treated and control (non-treated, matched) set to use in the difference-in-difference estimation.

Because the fee was only implemented on the derivatives market, the obvious choice for the non-treated control set is the underlying spot market. But there are strong linkages between the stock futures market and the underlying spot market. Higher cost of futures trading will make it more attractive for traders to automatically use the equity spot (Aggarwal and Thomas, 2011), in which case, we expect some trading to migrate to the spot market from the futures market. Therefore, we expect a corresponding but *indirect* impact of the fee on the underlying spot market. Further, because the derivatives and the spot market are connected by arbitrage, any adverse effect of the OTR fee on the derivatives could also affect the spot market quality adversely. These reasons lead us to expect an endogeneity bias in our estimated impact of the fee on market quality, which could also contaminate the inference when the underlying spot is used as a control for the futures.

To control for this, we propose a treated and control group where the control was not affected by the fee at all. Here, the treated group includes the stocks which have SSF trading while the control is selected from stocks which do not have SSF trading. At the NSE, derivatives are only traded on stocks which satisfy the following criteria:

1. The stock should be in the top 500 in terms of average daily market capitalisation **and** average daily traded value in the previous six months on a rolling basis.
2. The median ‘quarter-sigma order size’ for the stock should not be less than an average of Rs.1 million over the last six months.

3. The market wide position limit (determined by the number of shares held by non-promoters) in the stock should not be less than Rs.3 billion.

We use these requirements to identify non-SSF stocks that were close to the threshold, but did not satisfy them completely.<sup>7</sup> These non-SSF stocks constitute the *control* group, and the SSF stocks are the *treated* group.

Once we identify the treated and the control group, we estimate a difference-in-differences (DiD) regression to measure the impact of the fee, similar to the approach followed by Friederich and Payne (2015). The DiD is estimated using the following model specification:

$$\begin{aligned} \text{MEASURE}_{i,t} = & \alpha + \beta_1 \times \text{TREATED}_i + \beta_2 \times \text{FEE}_t + \\ & \beta_3 \times \text{TREATED}_i \times \text{FEE}_t + \\ & \beta_4 \times \text{MCAP}_{i,t} + \beta_5 \times \text{INVERSE-PRICE}_{i,t} + \\ & \beta_6 \times \text{NIFTY-VOL}_t + \beta_7 \times \text{ATINTENSITY}_t + \\ & \beta_8 \times \text{ROLLOVER-DUMMY} + \epsilon_{i,t} \end{aligned}$$

where,  $\text{MEASURE}_{i,t}$  is the OTR for a given stock in the SSF or any of the market quality measures described in Section 4.2.  $\text{TREATED}_i$  is a dummy variable which takes value 1 for a treated stock, 0 otherwise.  $\text{FEE}_t$  is a time dummy which takes value 1 for the period post the fee imposition, 0 otherwise. The estimated coefficient of interest is  $\hat{\beta}_3$  which measures the causal impact of the fee on  $\text{MEASURE}_{i,t}$ .

We also include control variables to account for variation caused by stock specific and macro-economic factors. For this, we use stock size ( $\text{MCAP}_{i,t}$ ) and relative tick size measured by the inverse of the stock price for stock variation, and market volatility for the macro-economy, which is measured as the realized volatility of intra-day returns on Nifty index ( $\text{NIFTY-VOL}$ ). We also control for the level of algorithmic trading by including the variable  $\text{AT-INTENSITY}$ , which measures the percentage of trades where there was an algorithmic trader on either one side or both side of the trade. To control for rollover effects on the SSF market from near month to next month expiry, we use a rollover dummy that is 1 two days prior to expiry, and 0 otherwise. All variables are winsorised at the 99% and 1% levels. The reported standard errors are clustered at the firm and time levels.

We estimate Equation 3.1 with the SSF as the treated (SSF-treated) group and *matched* spot as control (Spot-control) to measure the *direct* impact on the futures market.

We also estimate the equation with the stock with SSF as treated (Spot-treated) and *matched* stock without SSF as control (Spot-control) to measure the *indirect* impact on

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<sup>7</sup>This criteria brings us close to a regression discontinuity design (RDD). However, because of a lack of a clear definition of thresholds for market value and traded volume, we do not use the RDD framework.

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**Table 2** Number of stocks used in matching stocks with SSF and stocks without SSF

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The table shows the matched sample for Event 1 when the fee was implemented by the exchange, and Event 2 when the fee was imposed by the regulator.

‘Initial sample’ indicates the number of stocks in the treated and control groups before matching. ‘Final sample’ indicates the number of stocks in each group after matching.

Treated sample is the set of stocks with derivatives, and control sample is the set of stocks without derivatives on the NSE equity platform.

|         | <b>Event 1</b> |              | <b>Event 2</b> |              |
|---------|----------------|--------------|----------------|--------------|
|         | Initial sample | Final sample | Initial sample | Final sample |
| Treated | 156            | 39           | 187            | 41           |
| Control | 344            | 39           | 313            | 41           |

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the underlying spot market. We describe the methodology used to select the matched control stocks for the treated stocks in the following section.

### 3.2 Sample construction: treated spot and matched control

We use propensity score matching on a set of defined covariates to identify the set of treated (stocks with SSF) and control (stocks without SSF) groups. These covariates are based on the eligibility criteria used by NSE to select which stocks to trade derivatives on. These include market capitalisation, prices, floating stock, turnover and number of trades (Davies and Kim, 2009; Aggarwal and Thomas, 2014). We take the average value of each covariate for the period before the fee was announced, and estimate a propensity score using a logistic regression. We conduct a one-to-one matching on estimated propensity scores for each firm using the nearest neighbor matching algorithm (without replacement) and a caliper of 0.05.

Table 2 presents details of the initial sample<sup>8</sup> and the final sample used in the analysis for both the events. The final sample obtained after matching has 39 treated and control stocks for Event 1 and 41 treated and control stocks for Event 2. Figure 2 presents the empirical distribution of the propensity score of the two groups, before and after matching. The overlap between the density of the two sets before matching indicates the region of common support, which has a tight overlap after matching for both the events.<sup>9</sup>

Table 3 reports the match balance statistics for each event and shows that there is a good match balance across all matching covariates between the treated and control firms in the final sample *prior* to the intervention. We also achieve a near balance on outcome variables

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<sup>8</sup>We restrict the analysis to the top 500 stocks by market capitalisation. We also exclude stocks that underwent any corporate action including stock split, merger, rights and bonus issue or a buyback.

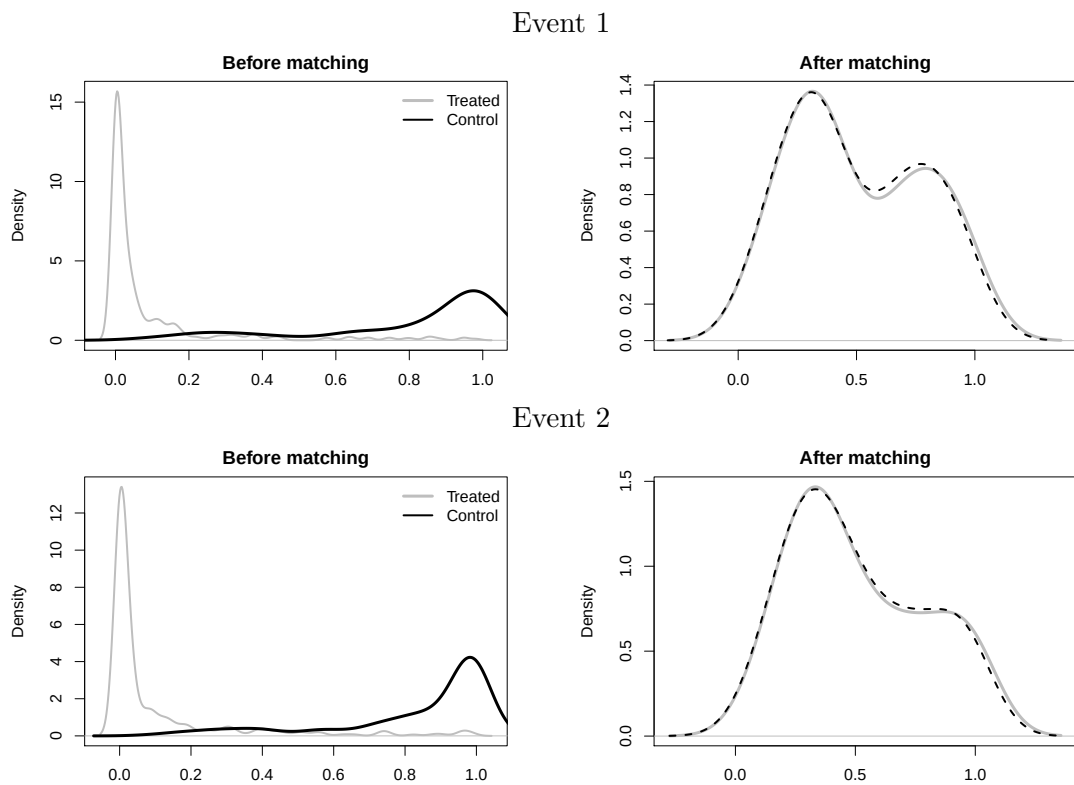
<sup>9</sup>The sample can be enlarged by using less stringent criteria in matching. For example, at a caliper of 0.25, the sample size results in 45 treated and control stocks for 2009 and in 46 treated and control stocks for 2012. However, we lose match balance based on that caliper size.

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**Figure 2** Empirical distribution of the propensity scores before and after matching

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The graphs show the density plot of the propensity score of the initial and final sample before and after matching for Events 1 and 2.





**Table 3** Match balance statistics for Event 1 and Event 2

The table provides match balance statistics for the matched sample for both the events prior to the fee implementation. Panel A shows the matched balance statistics for Event 1 and Panel B shows the statistics for Event 2.  $\mu_{tr}$  is the mean for the treated stocks, and  $\mu_{cr}$  is the mean for the control stocks. The p-value is reported based on the t-test and Kolmogorov-Smirnov test for equality of mean and distribution, respectively.

**Panel A: Event 1**

|                 | Before matching |            |              |      | After matching |            |              |      |
|-----------------|-----------------|------------|--------------|------|----------------|------------|--------------|------|
|                 | $\mu_{tr}$      | $\mu_{cr}$ | p-value<br>t | KS   | $\mu_{tr}$     | $\mu_{cr}$ | p-value<br>t | KS   |
| Distance (PS)   | 0.81            | 0.09       | 0.00         | 0.00 | 0.51           | 0.50       | 0.88         | 1.00 |
| ln(MCap)        | 11.33           | 9.31       | 0.00         | 0.00 | 10.34          | 10.34      | 0.23         | 0.75 |
| ln(Turnover)    | 5.88            | 2.79       | 0.00         | 0.00 | 4.87           | 4.88       | 0.44         | 0.56 |
| Floating stock  | 49.17           | 45.20      | 0.04         | 0.14 | 51.33          | 44.88      | 0.11         | 0.15 |
| ln(Price)       | 5.51            | 5.07       | 0.01         | 0.00 | 5.09           | 5.22       | 0.76         | 0.39 |
| ln(# of trades) | 9.76            | 7.24       | 0.00         | 0.00 | 9.06           | 0.96       | 0.75         |      |

**Panel B: Event 2**

|                 |       |       |      |      |       |       |      |      |
|-----------------|-------|-------|------|------|-------|-------|------|------|
| Distance (PS)   | 0.84  | 0.10  | 0.00 | 0.00 | 0.51  | 0.51  | 0.89 | 1.00 |
| ln(MCap)        | 11.35 | 9.76  | 0.00 | 0.00 | 10.82 | 10.52 | 0.07 | 0.42 |
| ln(Turnover)    | 5.30  | 2.09  | 0.00 | 0.00 | 4.15  | 4.16  | 0.29 | 0.99 |
| Floating stock  | 47.94 | 40.32 | 0.00 | 0.00 | 45.86 | 43.00 | 0.56 | 0.92 |
| ln(Price)       | 5.27  | 5.19  | 0.60 | 0.63 | 5.21  | 5.25  | 0.46 | 0.93 |
| ln(# of trades) | 9.52  | 6.70  | 0.00 | 0.00 | 8.57  | 8.56  | 0.41 | 0.59 |

of OTR and market quality based on our matching criteria.<sup>10</sup>

## 4 OTR and market quality measures

We next discuss the outcome variables on which we assess the impact of the OTR fee. The first step is to calculate the OTR for each stock.

### 4.1 OTR

We compute the OTR in two different ways: at an order level, and at the aggregate level, for a stock. To calculate the OTR at an order level, for each unique order for a stock on a day, we compute the following:

<sup>10</sup>Figure 7-10 in the appendix provides graphical evidence of the common trends assumption on the outcome variables. The t-test for the mean of the outcome variables for the treated and control sets does not reject the equality of means for these two sets prior to treatment for both the events.

$$OTR = \frac{\# \text{ of order events}}{1 + \# \text{ of trades}}$$

where the order events include entry, modifications and cancellations. For any given day, there may be several orders with multiple order events but no trades. To eliminate the problem of an infinite OTR value, we add 1 to the denominator of the ratio. We then take a value-weighted OTR average to arrive at a daily value. We call this the OTR VWTD.

For the OTR at an aggregate level, we calculate the ratio of the total number of messages for each stock for each day received by the exchange to the total number of trades. The number of messages includes order entry, modification and cancellation. We call this the OTR NAIVE.

Figures 3 and 4 shows the two OTR measures for the spot and the SSF markets for Event 1 by market-cap quantiles. We see that both the OTR measures declined on the SSF market after the fee was implemented across all market-cap quantiles. However, the OTR on the spot market increases following its pre-event trend. This indicates that the fee may not have had any impact on the underlying spot market.

Figures 5 and 6 shows the two OTR measures for the spot and the SSF markets for Event 2. Unlike for Event 1 described in the previous paragraphs, we do not see any impact on any of the measures either on the SSF market or on the spot market.

## 4.2 Market quality

The quality of the market is typically measured as available liquidity and the efficiency of its prices. Upon either of liquidity or efficiency increasing, so does the market quality. We use a few standard measures of liquidity and efficiency in this paper.

**Liquidity** We measure liquidity in two ways: as transactions costs, and as available depth.

*quoted spread*, *impact cost* and *ILLIQ* are three market transactions cost measures. QSPREAD captures the cost for a small order by examining the percentage difference between the ask and bid prices. IMPACT COST measures the instantaneous cost for a given quantity. We measure the impact cost at three order sizes: Rs. 250,000 (USD 3,800), Rs. 500,000 (USD 7,600) and Rs. 1,000,000 (USD 15,200). Although these transaction sizes are small by global standards, we use these because Rs.25,000 is the size of an average trade in the equity spot market and Rs.250,000 is the lot size in the derivatives market.<sup>11</sup>

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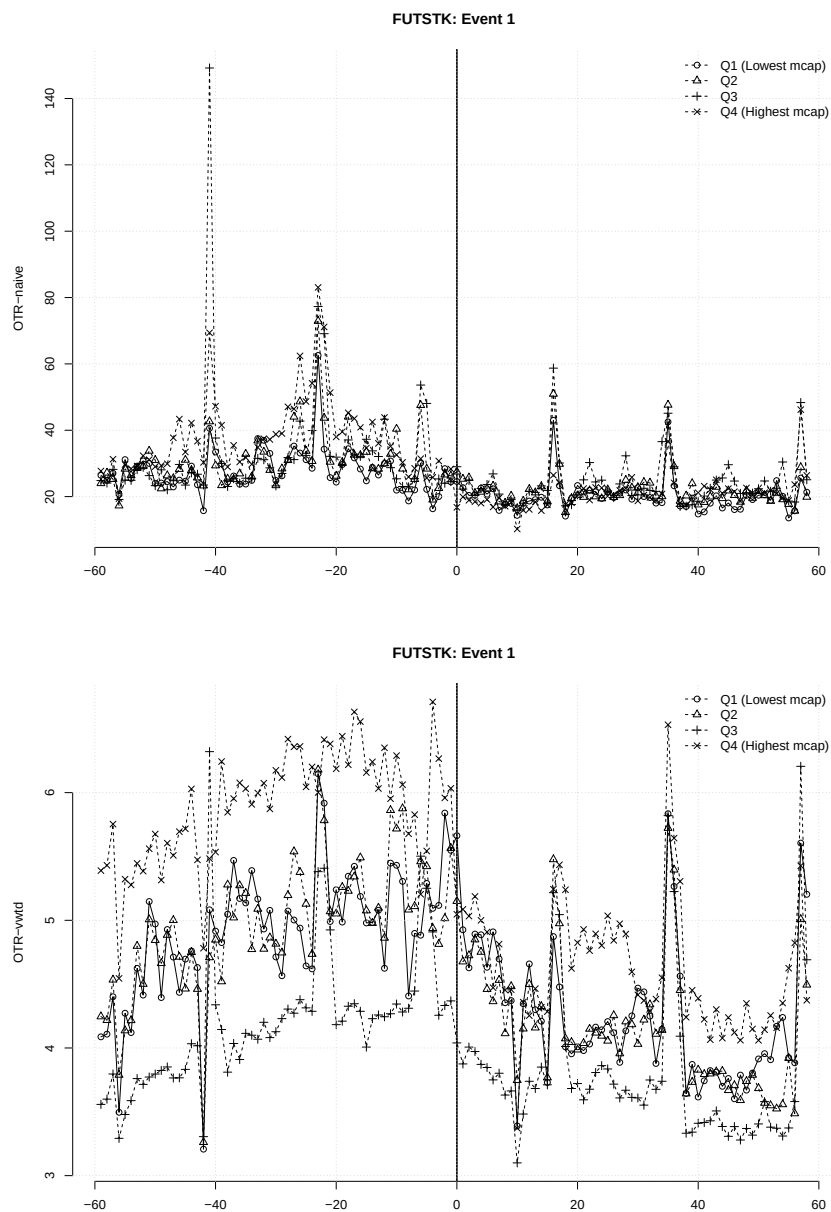
<sup>11</sup>Since the writing of this paper, the lot size in the derivatives markets has been increased to Rs.500,000 or approximately USD 7800, effective as of 28-04-2015. This is beyond the period of the analysis and does not affect our results.

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**Figure 3** OTR in SSF markets across size quartiles, Event 1

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The graphs below shows the market-cap weighted average daily OTR VWTD and OTR NAIVE on the SSF market for Event 1 for different market-cap based quartiles.

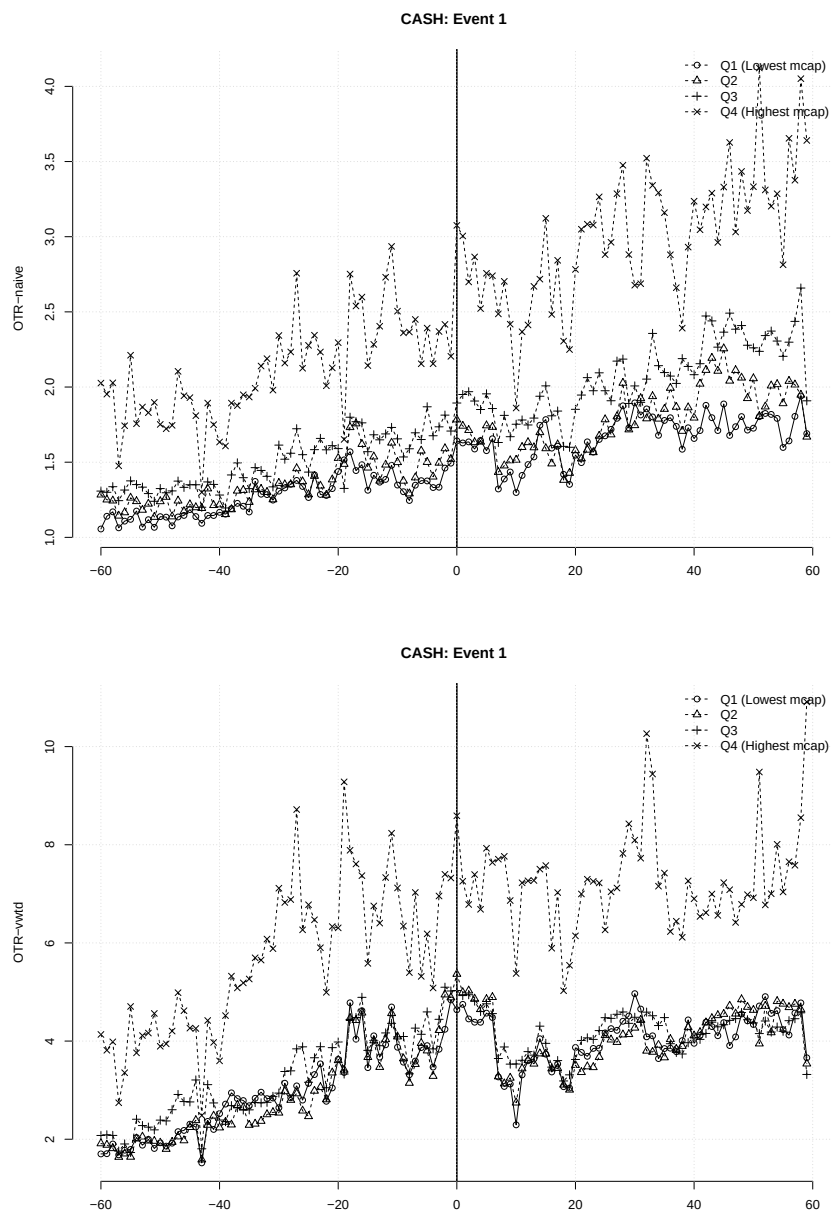


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**Figure 4** OTR for spot market across size quartiles, Event 1

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The graphs below shows market-cap weighted average daily OTR VWTD and OTR NAIVE on the spot market for Event 1 for different market-cap based quartiles.

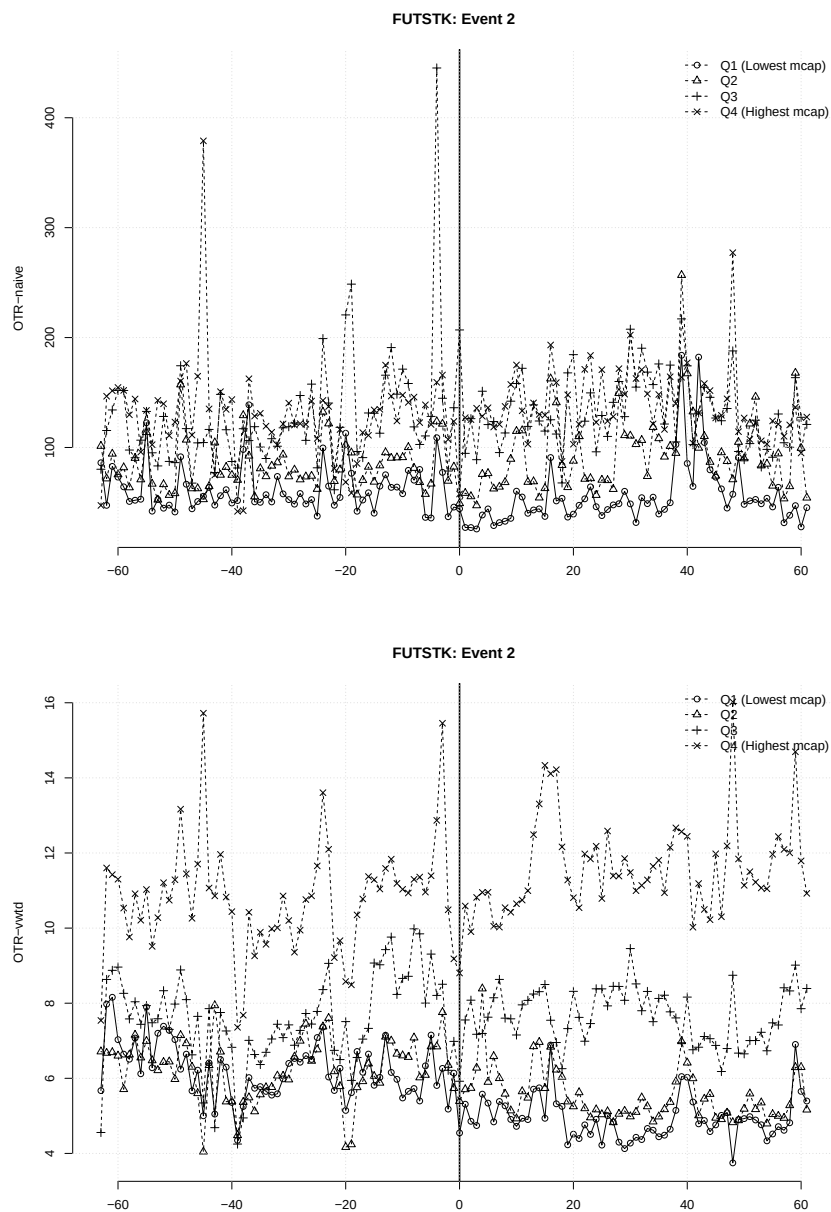


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**Figure 5** OTR of SSF markets across size quartiles, Event 2

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The graphs below shows market-cap weighted average daily OTR VWTD and OTR NAIVE on the SSF market for Event 2 for different market-cap based quartiles.

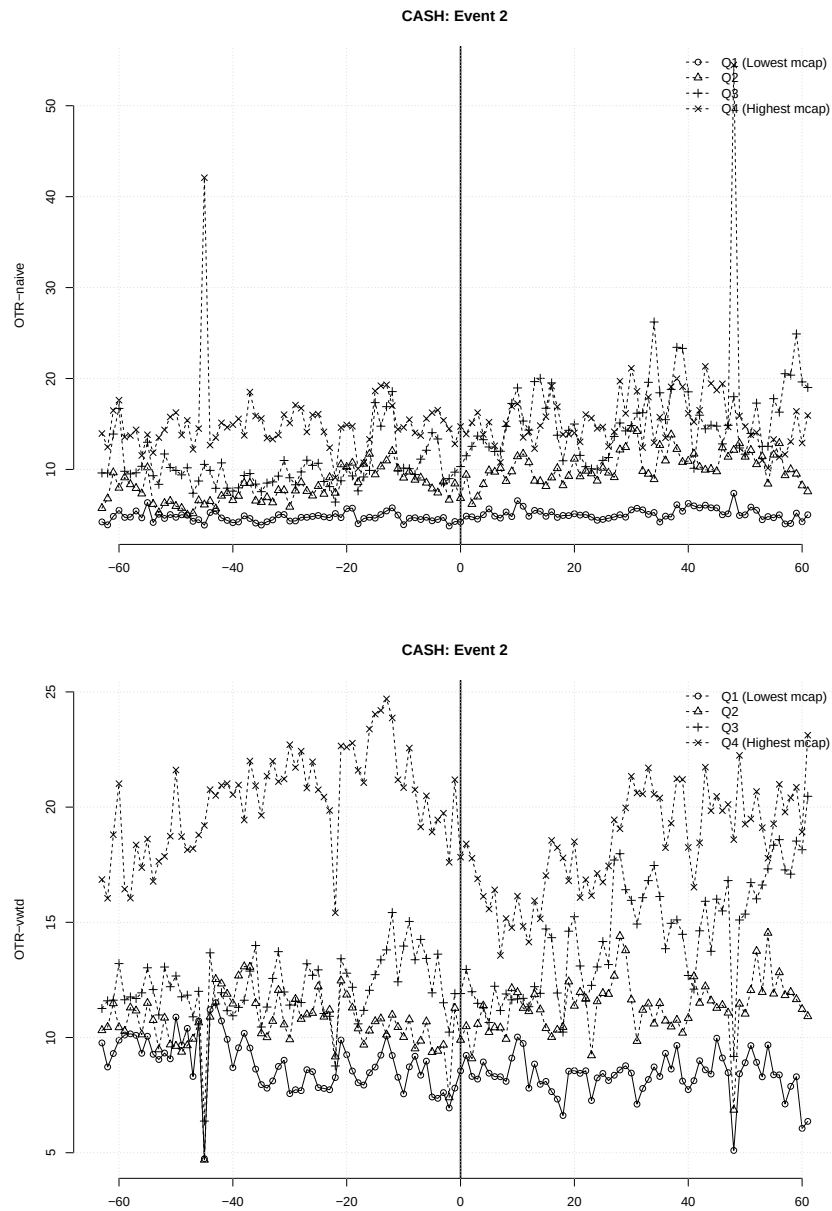


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**Figure 6** OTR for spot across size quartiles, Event 2

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The graphs below shows market-cap weighted average daily OTR VWTD and OTR NAIVE on the spot market for Event 2 for different market-cap based quartiles.



We additionally compute the Amihud illiquidity measure (*ILLIQ*) (Amihud, 2002) as the ratio of absolute returns to total traded value in a day.

For depth measures of liquidity, because the data allow us to construct the full limit order book, we calculate the following four measures: (1) the rupee value of orders available at the best prices in the limit order book (*TOP1DEPTH*), (2) the rupee value of orders available across the best five prices (or *TOP5DEPTH*), (3) the rupee value of orders available across the best seven prices (or *TOP7DEPTH*), and, (4) the rupee value of orders available across the best 10 prices (or *TOP10DEPTH*),

All the liquidity measures – other than *ILLIQ* – are first calculated for each stock at the frequency of a second, and the median value is used as the depth measure for the day. *ILLIQ* is calculated directly as a daily value for each stock and does not depend upon intra-day, high-frequency data.

**Efficiency** We measure informational efficiency of prices using two measures: the variance ratio of returns and liquidity risk.

The *variance ratio* or *VR* (Lo and MacKinlay, 1988) is computed as the ratio of the 10 minute returns variance to the 5 minute returns variance. A *VR* of 1 indicates a random walk. Under the null hypothesis of prices following a random walk,  $|VR - 1|$  should be zero.

The *volatility of liquidity* measures liquidity risk. An argument often made against *AT* is that it presents orders to the limit order book, but withdraws these orders before another trader can act on it. Such behaviour in the market implies that we should expect high liquidity risk when there is high *AT* on the stock. We use the intra-day limit order book information to calculate the impact cost as market liquidity and the standard deviation of the impact cost as the volatility of liquidity or *LIQRISK*.

As with the liquidity measures, the efficiency measures are also calculated using intra-day, high-frequency data from the NSE. In addition to these, we also report *realised volatility* (or  $\sigma_r$ ) for each stock, which is calculated as the standard deviation of 5 minute returns.

## 5 Results

In this section, we present the results of the equation described in Section 3. The results are ordered by the impact of the fee on the *OTR* described in Section 4.1 and the causal impact of the fee on market quality described in Section 4.2.

**Table 4** Difference-in-Difference estimates for impact on the OTR, Event 1

The table reports the results of the DiD regression to assess the impact of the OTR fee charged by the exchange (Event 1) on OTR VWTD and OTR NAIVE.

‘Fee’ is the variable differentiating treated and control, while ‘Treated’ differentiates the pre-OTR fee and post-OTR fee periods. ‘Treated  $\times$  Fee’ is the interaction term that captures the estimated treatment effect of the fee on the OTR for the treated sample.

The  $t$ -statistics based on standard errors clustered by stock and time are provided in parentheses. **Boldface** values indicate significance at 5%.

|                      | SSF(treated)-Spot(control) |                         | Spot(treated)-Spot(control) |                         |
|----------------------|----------------------------|-------------------------|-----------------------------|-------------------------|
|                      | OTR VWTD                   | OTR NAIVE               | OTR VWTD                    | OTR NAIVE               |
| Fee                  | -0.08<br>(-1.64)           | <b>-0.42</b><br>(-2.09) | <b>-0.16</b><br>(-2.67)     | 0.04<br>(1.71)          |
| Treated              | <b>2.88</b><br>(11.14)     | <b>22.36</b><br>(15.11) | <b>1.33</b><br>(7.77)       | <b>0.24</b><br>(3.88)   |
| Treated $\times$ Fee | 0.07<br>(0.49)             | <b>-3.45</b><br>(-3.19) | <b>1.5</b><br>(6.78)        | <b>0.32</b><br>(5.61)   |
| Market Cap           | 0.22<br>(1.73)             | -0.39<br>(-0.72)        | 0.29<br>(1.12)              | 0.04<br>(0.8)           |
| Inverse Price        | <b>0.08</b><br>(2.81)      | 0.11<br>(1.67)          | -0.01<br>(-0.69)            | <b>-0.02</b><br>(-4.42) |
| Market Vol           | <b>-0.02</b><br>(-2.79)    | -0.03<br>(-1.23)        | <b>-0.02</b><br>(-4.37)     | <b>0</b><br>(-2.47)     |
| AT Intensity         | <b>0.02</b><br>(2.54)      | <b>0.24</b><br>(4.72)   | <b>0.02</b><br>(2.72)       | <b>0</b><br>(2)         |
| Rollover             | <b>0.25</b><br>(2.58)      | <b>5.02</b><br>(4.05)   | -0.08<br>(-1.43)            | 0.01<br>(0.33)          |
| Adj. R <sup>2</sup>  | 0.64                       | 0.65                    | 0.45                        | 0.34                    |
| Treated              | 39                         | 39                      | 39                          | 39                      |
| Control              | 39                         | 39                      | 39                          | 39                      |
| # of obs.            | 6060                       | 6060                    | 6715                        | 6715                    |

## 5.1 Impact on OTR

Table 4 presents the estimation results for Event 1, when the fee was imposed by NSE in 2009. Table 5 presents the estimation results for Event 2 when the fee was imposed by SEBI in 2012.

### Event 1

The results show that OTR NAIVE *reduced* for the treated stocks on the SSF market relative to the control stocks on the spot market (first two columns, Table 4). OTR NAIVE for the matched treated SSF stocks declined 3.45 times in magnitude, relative to the matched



control spot stocks after the imposition of the fee. We do not see any significant impact on OTR VWTD. This result can be attributed to the objective of the fee. Because the fee was primarily targeted to curb the high rate of IOC orders in the market, which have an OTR of 1 by definition, we do not see any impact on VWTD OTR.

The last two columns of Table 4 show that the OTR on the spot market for the treated stocks *increased* relative to the control stocks. This is true for both, value-weighted OTR captured at an individual order level as well as OTR captured an aggregate level. We acknowledge that the OTR VWTD was already on an increasing trend on the spot market for the treated stocks but not for the control stocks (Figure 8). However, this is not true of OTR NAIVE where both the matched treated and control stocks were on a similar trajectory prior to the intervention. The estimated coefficient for the interaction term for OTR NAIVE is positive and significant, implying an increase in the OTR NAIVE 0.32 times that on the spot market. This suggests that the fee imposed by the exchange resulted in the migration of some high OTR-based trading from SSF to the underlying spot market.

In summary, the results indicate that the fee imposed by the exchange resulted in a decline in aggregate OTR on the SSF market and an increase in the aggregate OTR of the spot market. This shows that the fee was effective in curbing the high OTR on the SSF market.

## Event 2

The fee in 2012 was only levied on *algorithmic* orders with exemptions (as described in Section 2.2). In addition to the specification in Equation 3.1, we include an exclusion dummy our DiD estimation to control for the effect of the exclusion of stocks from the derivative segment.<sup>12</sup> The dummy is 1 for excluded stocks from the date of announcement, 0 otherwise.

Table 5 reports a positive and significant coefficient on the ‘Treated’ dummy variable for OTR NAIVE and OTR VWTD. However, the coefficient of the interaction term ‘Treated  $\times$  Fee’ is not significantly different from 0 for OTR NAIVE, which shows that the fee did not have any impact on the overall level of OTR. This is true for treated stocks compared with controls, whether it is the SSF versus the spot, on just the spot with derivatives stocks versus not. We highlight the caveat that the common trends assumption is violated for OTR VWTD, based on Figure 9.

These results indicate that there was no impact of the 2012 fee on the OTR, contrary to what we expect from an additional friction on high frequency trading. This is *especially* so because there was a significantly higher fraction of AT in the markets in 2012 compared to 2009, when the OTR FEE was effective in bringing down the market-wide OTR levels.

One explanation could be that because the fee was imposed in an asymmetric manner across

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<sup>12</sup>In June 2012, 26 of the 41 treated stocks were announced by the exchange as being excluded from the derivatives segment, as of September 2012.

**Table 5** Difference-in-Difference estimates for impact on the OTR, Event 2

The table reports the DiD estimation results for the impact of the OTR fee charged by the regulator (Event 2) on OTR VWTD and OTR NAIVE.

‘Fee’ is the variable differentiating treated and control, while ‘Treated’ differentiates the pre-OTR fee and post-OTR fee periods. ‘Treated  $\times$  Fee’ is the interaction term that captures the causal effect of the fee on the OTR for the treated sample.

The  $t$ -statistics based on standard errors clustered by stock and time are provided in parentheses. **Boldface** values indicate significance at 5%.

|                         | SSF(treated)-Spot(control) |                        | Spot(treated)-Spot(control) |                         |
|-------------------------|----------------------------|------------------------|-----------------------------|-------------------------|
|                         | OTR VWTD                   | OTR NAIVE              | OTR VWTD                    | OTR NAIVE               |
| Fee                     | <b>1.00</b><br>(3.1)       | <b>2.87</b><br>(3.19)  | <b>1.06</b><br>(3.2)        | <b>1.47</b><br>(3.32)   |
| Treated                 | 0.45<br>(0.87)             | <b>60.69</b><br>(8.69) | <b>4.9</b><br>(5.04)        | 1.31<br>(0.85)          |
| Treated $\times$ Fee    | <b>-1.37</b><br>(-2.77)    | 7.41<br>(0.63)         | 1.91<br>(1.35)              | 4.42<br>(1.49)          |
| Market Cap              | 0.43<br>(1.56)             | 0.19<br>(0.08)         | 0.14<br>(0.41)              | 0.74<br>(1.22)          |
| Inverse Price           | <b>-0.03</b><br>(-2.77)    | -0.17<br>(-1.78)       | <b>-0.06</b><br>(-3.19)     | <b>-0.11</b><br>(-3.09) |
| Market Vol              | <b>-0.02</b><br>(-2.97)    | 0.24<br>(1.86)         | 0<br>(0.27)                 | -0.02<br>(-1.57)        |
| AT Intensity            | <b>0.03</b><br>(2.89)      | 0.09<br>(0.59)         | 0.02<br>(0.96)              | -0.04<br>(-1.27)        |
| Rollover                | -0.04<br>(-0.17)           | 0.73<br>(0.63)         | <b>-0.46</b><br>(-2.58)     | 0.61<br>(1.89)          |
| Excluded                | -0.34<br>(-0.7)            | -3.48<br>(-0.23)       | <b>-4.19</b><br>(-2.19)     | -6.36<br>(-1.63)        |
| Adjusted R <sup>2</sup> | 0.11                       | 0.26                   | 0.23                        | 0.13                    |
| Treated                 | 41                         | 41                     | 41                          | 41                      |
| Control                 | 41                         | 41                     | 41                          | 41                      |
| # of obs                | 7485                       | 7485                   | 9515                        | 9515                    |

orders in the book, traders reduced orders where the fee was binding while increasing orders where the fee was not applicable. To test this hypothesis, we examine the percentage of orders entered beyond one percent (ORDERS-BEYOND) price limit for our sample of treated and matched control stocks (results shown in Table 6).

We find a significant *reduction* in the percentage of orders placed in the limit order book for treated stocks, beyond the one percent price limit where the OTR FEE became binding. However, we see a similar effect in the underlying market for the matched treated stocks, i.e., the matched control stocks. Thus, the fee appears to have been effective in reducing OTR on the SSF market, with a simultaneous impact on the underlying market. However, whether this was an optimal outcome for market quality remains to be tested. The decline in the percentage of orders beyond the one percent price limit could have an implication for the overall depth of the market as well as the liquidity available in the market, at prices away from the best bid and ask price.

## 5.2 Impact on liquidity

We next examine the impact of the fee on market liquidity. Table 7 presents the DiD estimation results for the effect of the OTR fee on the liquidity variables for Event 1 described in Section 4.2, and Table 8 presents the results for Event 2. In both cases, Panel A presents the results for the direct impact on the SSF market, where the SSF is the matched treated set relative to the matched control set on the underlying spot market. Panel B presents the results for the indirect impact on the spot market, where the stocks with the SSF are the treated stocks and the matched stocks without the SSF are the control.

### Event 1

The estimated coefficient of the interaction term is significant across all liquidity measures in Panel A. The coefficients are negative and significant for the transactions costs measures (QSPREAD and IMPACT COST at different transaction size). This indicates that there was decline in transactions costs on the SSF market for treated stocks after the OTR fee was imposed in Event 1. Quoted spread declined by 6 basis points for the treated stocks on the SSF market relative to the matched control stocks on the spot market. The impact cost declined by 3 to 10 basis points at different transaction sizes level. The coefficients of the depth measures are also positive and significant implying an improvement in depth after the implementation of the fee. Depth improved by 13-15 percent across different levels. The negative and significant coefficient of the Amihud's illiquidity measure confirms that liquidity for treated stocks improved on the SSF market relative to the spot market.

Panel B of Table 7 shows that the OTR fee did *not* have a significant impact on most of the transactions costs measures, but did have a positive impact on the depth for the

**Table 6** Difference-in-Difference estimates for impact of the fee on the percentage of orders entered beyond 1% after Event 2

The table reports the DiD estimation results for the impact of the OTR fee charged by the regulator (Event 2) on the percentage of orders entered beyond 1 percent (ORDERS-BEYOND).

‘Fee’ is the variable differentiating treated and control while ‘Treated’ differentiates the pre-OTR fee and post-OTR fee periods. ‘Treated  $\times$  Fee’ is the interaction term that captures the causal effect of the fee on the OTR for the treated sample.

The  $t$ -statistics based on standard errors clustered by stock and time are presented in parentheses. **Boldface** values indicate significance at 5%.

|                         | SSF(treated)-Spot(control)<br>ORDERS-BEYOND | Spot(treated)-Spot(control)<br>ORDERS-BEYOND |
|-------------------------|---------------------------------------------|----------------------------------------------|
| Fee                     | -2.67<br>(-1.81)                            | <b>-3.47</b><br>(-2.36)                      |
| Treated                 | -3.46<br>(-1)                               | <b>11.42</b><br>(3.68)                       |
| Treated $\times$ Fee    | <b>-12.18</b><br>(-4.09)                    | <b>-7.01</b><br>(-2.63)                      |
| Market Cap              | 0.18<br>(0.15)                              | 0.69<br>(0.61)                               |
| Inverse Price           | <b>0.32</b><br>(3.81)                       | <b>0.39</b><br>(3.81)                        |
| Market Vol              | 0.03<br>(0.34)                              | <b>-0.14</b><br>(-3.15)                      |
| AT Intensity            | <b>-0.33</b><br>(-5.93)                     | <b>-0.38</b><br>(-8)                         |
| Rollover                | 0.98<br>(1.01)                              | <b>1.65</b><br>(2.41)                        |
| Excluded                | <b>11.77</b><br>(2.54)                      | <b>9.36</b><br>(2.62)                        |
| Adjusted R <sup>2</sup> | 0.22                                        | 0.30                                         |
| Treated                 | 41                                          | 41                                           |
| Control                 | 41                                          | 41                                           |
| # of obs                | 7485                                        | 9514                                         |

**Table 7** Difference-in-Difference estimates for impact on market liquidity, Event 1

The table reports the results of daily panel DiD estimations on the liquidity variables: QSPREAD, and  $IC_{250k}$ ,  $IC_{500k}$ ,  $IC_{1000k}$ , TOP1DEPTH, TOP5DEPTH, TOP7DEPTH, TOP10DEPTH and ILLIQ. Panel A presents the results for matched treated stocks on SSF market and the matched control stocks on the spot market. Panel B presents the results for matched treated and controls sets on the spot market.

‘Fee’ is the variable differentiating treated and control, while ‘Treated’ differentiates the pre-OTR fee and post-OTR fee periods. ‘Treated  $\times$  Fee’ is the interaction term that captures the impact of the fee on the OTR for the treated sample.

The  $t$ -statistics based on standard errors clustered by stock and time are presented in parentheses. **Boldface** values indicate significance at 5%.

|                                        | QSPREAD                | $IC_{250k}$             | $IC_{500k}$             | $IC_{1000k}$            | TOP1DEPTH                | TOP5DEPTH                | TOP7DEPTH               | TOP10DEPTH              | ILLIQ                   |
|----------------------------------------|------------------------|-------------------------|-------------------------|-------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------------------|
| Panel A: SSF (treated) -Spot(control)  |                        |                         |                         |                         |                          |                          |                         |                         |                         |
| Fee                                    | 0.01<br>(1.91)         | -0.01<br>(-1.94)        | -0.02<br>(-1.76)        | 0<br>(0)                | 0.03<br>(0.82)           | 0.04<br>(1.16)           | 0.05<br>(1.26)          | 0.04<br>(1.06)          | -0.24<br>(-0.58)        |
| Treated                                | <b>0.13</b><br>(9.48)  | -0.02<br>(-1.19)        | -0.04<br>(-1.79)        | 0.03<br>(1.05)          | <b>1.9</b><br>(19.11)    | <b>1.69</b><br>(18.83)   | <b>1.68</b><br>(18.77)  | <b>1.66</b><br>(18.81)  | -1.22<br>(-1.49)        |
| Treated $\times$ Fee                   | <b>-0.06</b><br>(-6.8) | <b>-0.03</b><br>(-2.71) | <b>-0.05</b><br>(-3.41) | <b>-0.1</b><br>(-5.79)  | <b>0.13</b><br>(2.53)    | <b>0.15</b><br>(2.59)    | <b>0.14</b><br>(2.49)   | <b>0.14</b><br>(2.51)   | <b>-1.18</b><br>(-2.08) |
| Market Cap                             | -0.01<br>(-1.86)       | -0.02<br>(-1.77)        | -0.02<br>(-1.87)        | <b>-0.03</b><br>(-2.19) | 0.05<br>(1.12)           | 0.06<br>(1.25)           | 0.06<br>(1.16)          | 0.05<br>(1.02)          | -0.54<br>(-1.25)        |
| Inverse Price                          | 0<br>(-0.78)           | 0<br>(-1.22)            | 0<br>(-0.1)             | 0<br>(1.07)             | <b>0.02</b><br>(2.26)    | <b>0.03</b><br>(3.83)    | <b>0.03</b><br>(3.71)   | <b>0.03</b><br>(3.63)   | -0.04<br>(-0.93)        |
| Market Vol                             | <b>0</b><br>(9.35)     | <b>0</b><br>(13.52)     | <b>0</b><br>(13.23)     | <b>0.01</b><br>(9.82)   | <b>-0.01</b><br>(-7.68)  | <b>-0.01</b><br>(-10.06) | <b>-0.01</b><br>(-9.84) | <b>-0.01</b><br>(-9.97) | <b>0.16</b><br>(7.73)   |
| AT Intensity                           | 0<br>(-1.56)           | 0<br>(-1.63)            | 0<br>(-0.67)            | 0<br>(0.19)             | 0<br>(-0.61)             | 0<br>(-0.87)             | 0<br>(-0.95)            | 0<br>(-0.9)             | 0<br>(-0.16)            |
| Rollover                               | <b>0.01</b><br>(2.38)  | 0<br>(-0.19)            | -0.01<br>(-0.89)        | 0<br>(0.03)             | <b>0.12</b><br>(5.13)    | <b>0.13</b><br>(4.99)    | <b>0.13</b><br>(4.67)   | <b>0.13</b><br>(4.3)    | -0.17<br>(-0.33)        |
| Adjusted R <sup>2</sup>                | 0.46                   | 0.18                    | 0.19                    | 0.17                    | 0.83                     | 0.81                     | 0.8                     | 0.8                     | 0.06                    |
| Treated                                | 39                     | 39                      | 39                      | 39                      | 39                       | 39                       | 39                      | 39                      | 39                      |
| Control                                | 39                     | 39                      | 39                      | 39                      | 39                       | 39                       | 39                      | 39                      | 39                      |
| # of obs                               | 6060                   | 6058                    | 6037                    | 5740                    | 6060                     | 6060                     | 6060                    | 6060                    | 6060                    |
| Panel B: Spot (treated) -Spot(control) |                        |                         |                         |                         |                          |                          |                         |                         |                         |
| Fee                                    | 0<br>(-1.03)           | <b>-0.02</b><br>(-2.22) | -0.02<br>(-1.96)        | -0.01<br>(-0.55)        | -0.02<br>(-0.54)         | 0.01<br>(0.27)           | 0.01<br>(0.37)          | 0.01<br>(0.13)          | -0.3<br>(-0.74)         |
| Treated                                | -0.01<br>(-1.9)        | <b>-0.07</b><br>(-3.51) | <b>-0.07</b><br>(-3.31) | <b>-0.04</b><br>(-2.06) | <b>0.38</b><br>(4.36)    | <b>0.39</b><br>(4.66)    | <b>0.35</b><br>(4.16)   | <b>0.31</b><br>(3.63)   | <b>-2.1</b><br>(-2.79)  |
| Treated $\times$ Fee                   | 0<br>(0.7)             | 0.01<br>(0.94)          | 0.01<br>(0.52)          | -0.01<br>(-0.62)        | <b>0.19</b><br>(3.9)     | <b>0.18</b><br>(3.5)     | <b>0.19</b><br>(3.52)   | <b>0.21</b><br>(3.71)   | 0.36<br>(0.74)          |
| Market Cap                             | 0<br>(-0.65)           | -0.02<br>(-1.85)        | <b>-0.02</b><br>(-2.05) | <b>-0.03</b><br>(-2.28) | <b>0.17</b><br>(2.28)    | <b>0.15</b><br>(2.12)    | <b>0.15</b><br>(2.06)   | 0.15<br>(1.96)          | -0.48<br>(-1.27)        |
| Inverse Price                          | 0<br>(0.52)            | 0<br>(-0.25)            | 0<br>(0.72)             | <b>0</b><br>(2.14)      | <b>0.04</b><br>(5.36)    | <b>0.04</b><br>(7.27)    | <b>0.04</b><br>(7.33)   | <b>0.04</b><br>(7.4)    | -0.03<br>(-0.81)        |
| Market Vol                             | <b>0</b><br>(9.56)     | <b>0</b><br>(12.86)     | <b>0</b><br>(13.01)     | <b>0</b><br>(9.5)       | <b>-0.01</b><br>(-10.29) | <b>-0.02</b><br>(-10.05) | <b>-0.02</b><br>(-9.7)  | <b>-0.02</b><br>(-9.81) | <b>0.14</b><br>(7.29)   |
| AT Intensity                           | <b>0</b><br>(-3.46)    | <b>0</b><br>(-2.31)     | 0<br>(-1.78)            | 0<br>(-1.32)            | <b>0.01</b><br>(2.94)    | <b>0.01</b><br>(2.46)    | <b>0.01</b><br>(2.29)   | <b>0.01</b><br>(2.29)   | -0.02<br>(-1.11)        |
| Rollover                               | 0<br>(-1.41)           | <b>-0.01</b><br>(-6.39) | <b>-0.02</b><br>(-5.37) | <b>-0.02</b><br>(-3.69) | <b>0.09</b><br>(6.26)    | <b>0.09</b><br>(5.16)    | <b>0.09</b><br>(4.72)   | <b>0.09</b><br>(4.19)   | -0.44<br>(-1.43)        |
| Adjusted R <sup>2</sup>                | 0.1                    | 0.21                    | 0.19                    | 0.16                    | 0.48                     | 0.49                     | 0.46                    | 0.43                    | 0.06                    |
| Treated                                | 39                     | 39                      | 39                      | 39                      | 39                       | 39                       | 39                      | 39                      | 39                      |
| Control                                | 39                     | 39                      | 39                      | 39                      | 39                       | 39                       | 39                      | 39                      | 39                      |
| # of obs                               | 6715                   | 6713                    | 6692                    | 6379                    | 6715                     | 6715                     | 6715                    | 6715                    | 6715                    |

treated stocks relative to the matched control stocks. Depth on the spot market increased by 18 to 21 percent for the treated stocks relative to the matched control stocks after the event. Overall, these results show that market liquidity *benefited* from the OTR fee, both directly as well as indirectly.

## Event 2

Table 8 presents the DiD results of the impact of the OTR fee imposed by SEBI (Event 2) on liquidity. In Panel A, the significant coefficients of the interaction term ‘Treated $\times$ Fee’ are negative for quoted spread.

However, we need to be careful in interpreting the coefficient with quoted spread, as the transaction size of the quoted spread on the spot market is 10 times lower than the transaction size of the quoted spread on the SSF market. Transactions costs at higher quantities are not affected by the fee except at Rs.10 million, when observed at a 10% significance level. This finding also holds for the underlying spot market. In Panel B, we observe a negative and significant coefficient on the impact cost at Rs.10 million. This suggests an improvement in the liquidity of both the SSF and the underlying stock for the larger trade sizes after the fee. Panel B also indicates some evidence of improvement in the depth of liquidity for the treated stocks in the spot market as a consequence of the OTR fee.

The analysis suggests that liquidity in the markets actually benefits from the OTR fee. This was more clearly evident when the exchange imposed the fee to better manage their bandwidth and access to their broker members in 2009. The impact of the SEBI intervention is more difficult to identify. There is little impact on the OTR except where the fee is binding, and there is some mixed evidence indicating that there was an improvement in liquidity shown by depth in the market as a consequence of the 2012 fee.

## 5.3 Impact on efficiency

Table 9 presents the DiD results for the effect of the OTR fee on the efficiency measures described in Section 4.2 for Event 1, and Table 10 presents the estimation results for Event 2. Panel A presents the results for the direct impact on the SSF market, where the SSF is the matched treated set relative to the matched control set on the underlying spot market. Panel B presents the results for the indirect impact on the spot market, where the stocks with the SSF are the treated stocks and the matched stocks without the SSF are the control.

**Table 8** Difference-in-Difference estimates for impact on market liquidity, Event 2

The table reports the results of daily panel DiD estimations on the liquidity variables: QSPREAD, IC<sub>250k</sub>, IC<sub>500k</sub>, IC<sub>1000k</sub>, TOP1DEPTH, TOP5DEPTH, TOP7DEPTH, TOP10DEPTH and ILLIQ.

Panel A presents the results for treated stocks on SSF market versus the matched control stocks on the spot market. Panel B presents the results for matched treated and controls sets on the spot market.

‘Fee’ is the variable differentiating treated and control, while ‘Treated’ differentiates the pre-OTR fee and post-OTR fee periods. ‘Treated  $\times$  Fee’ is the interaction term that captures the impact of the fee on the OTR for the treated sample.

The  $t$ -statistics based on standard errors clustered by stock and time are presented in parentheses. **Boldface** values indicate significance at 5%.

|                                        | QSPREAD                 | IC <sub>250k</sub>      | IC <sub>500k</sub>      | IC <sub>1000k</sub>     | TOP1DEPTH               | TOP5DEPTH               | TOP7DEPTH               | TOP10DEPTH              | ILLIQ                   |
|----------------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Panel A: SSF (treated) -Spot(control)  |                         |                         |                         |                         |                         |                         |                         |                         |                         |
| Fee                                    | <b>-0.01</b><br>(-2.81) | <b>-0.03</b><br>(-4.07) | <b>-0.04</b><br>(-3.82) | <b>-0.03</b><br>(-2.17) | 0.09<br>(1.76)          | 0.1<br>(1.88)           | <b>0.11</b><br>(2.05)   | <b>0.12</b><br>(2.21)   | <b>-0.96</b><br>(-2.66) |
| Treated                                | <b>0.11</b><br>(8.7)    | <b>-0.04</b><br>(-2.22) | <b>-0.05</b><br>(-2.01) | 0.02<br>(0.69)          | <b>2.12</b><br>(16.78)  | <b>1.8</b><br>(14.23)   | <b>1.76</b><br>(13.79)  | <b>1.75</b><br>(13.82)  | <b>-1.76</b><br>(-2.63) |
| Treated $\times$ Fee                   | <b>-0.04</b><br>(-3.2)  | -0.01<br>(-0.46)        | -0.01<br>(-0.76)        | -0.06<br>(-1.92)        | 0.09<br>(1.04)          | 0.14<br>(1.4)           | 0.12<br>(1.25)          | 0.1<br>(1.05)           | 0.09<br>(0.16)          |
| Market cap                             | -0.01<br>(-1.55)        | <b>-0.02</b><br>(-3.89) | <b>-0.02</b><br>(-3.2)  | -0.02<br>(-1.66)        | <b>0.15</b><br>(3.45)   | <b>0.13</b><br>(2.26)   | <b>0.14</b><br>(2.31)   | <b>0.15</b><br>(2.34)   | <b>-0.68</b><br>(-3.14) |
| Inverse Price                          | <b>0</b><br>(7.94)      | <b>0</b><br>(2.79)      | <b>0</b><br>(2.59)      | <b>0.01</b><br>(2.8)    | <b>0.03</b><br>(4.57)   | <b>0.02</b><br>(3.98)   | <b>0.02</b><br>(3.67)   | <b>0.02</b><br>(3.42)   | 0.07<br>(1.91)          |
| Market Vol                             | <b>0</b><br>(5.96)      | <b>0</b><br>(6.29)      | <b>0</b><br>(4.51)      | <b>0</b><br>(4.15)      | <b>0</b><br>(-3.92)     | <b>0</b><br>(-4.17)     | <b>0</b><br>(-3.8)      | <b>0</b><br>(-3.74)     | <b>0.03</b><br>(2.5)    |
| AT intensity                           | <b>0</b><br>(-3.94)     | <b>0</b><br>(-2.11)     | <b>0</b><br>(-2.17)     | <b>0</b><br>(-1.98)     | 0<br>(-0.84)            | 0<br>(-1.21)            | 0<br>(-1.2)             | 0<br>(-1.31)            | -0.01<br>(-1.15)        |
| Rollover                               | 0<br>(-1.82)            | -0.01<br>(-1.16)        | -0.01<br>(-1.58)        | -0.03<br>(-1.62)        | 0<br>(-0.06)            | -0.02<br>(-0.45)        | -0.01<br>(-0.24)        | -0.02<br>(-0.47)        | -0.36<br>(-0.94)        |
| Excluded                               | <b>0.05</b><br>(3)      | 0.04<br>(1.77)          | <b>0.06</b><br>(2.03)   | <b>0.1</b><br>(2.27)    | -0.25<br>(-1.68)        | <b>-0.32</b><br>(-2.01) | -0.31<br>(-1.94)        | -0.27<br>(-1.74)        | 1.23<br>(1.34)          |
| Adjusted R <sup>2</sup>                | 0.56                    | 0.32                    | 0.3                     | 0.34                    | 0.76                    | 0.67                    | 0.65                    | 0.65                    | 0.11                    |
| Treated                                | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      |
| Control                                | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      |
| # of obs.                              | 7485                    | 7482                    | 7408                    | 6442                    | 7485                    | 7485                    | 7485                    | 7485                    | 7485                    |
| Panel B: Spot (treated) -Spot(control) |                         |                         |                         |                         |                         |                         |                         |                         |                         |
| Fee                                    | <b>-0.01</b><br>(-2.6)  | <b>-0.03</b><br>(-3.54) | <b>-0.03</b><br>(-3.29) | -0.02<br>(-1.66)        | 0.08<br>(1.59)          | 0.1<br>(1.8)            | <b>0.11</b><br>(1.98)   | <b>0.12</b><br>(2.16)   | <b>-0.96</b><br>(-2.64) |
| Treated                                | 0<br>(-0.17)            | -0.01<br>(-0.88)        | 0<br>(0.13)             | <b>0.08</b><br>(2.22)   | <b>0.32</b><br>(3.03)   | <b>0.34</b><br>(3.06)   | <b>0.34</b><br>(2.99)   | <b>0.35</b><br>(3.08)   | -0.57<br>(-0.83)        |
| Treated $\times$ Fee                   | 0<br>(-1.29)            | -0.02<br>(-1.39)        | -0.03<br>(-1.87)        | <b>-0.06</b><br>(-2.04) | <b>0.19</b><br>(2.17)   | 0.18<br>(1.87)          | 0.19<br>(1.93)          | 0.19<br>(1.95)          | -0.24<br>(-0.5)         |
| Market Cap                             | 0<br>(-1.62)            | <b>-0.03</b><br>(-4.6)  | <b>-0.03</b><br>(-4.14) | -0.03<br>(-1.87)        | <b>0.2</b><br>(3.03)    | <b>0.18</b><br>(2.32)   | <b>0.18</b><br>(2.34)   | <b>0.18</b><br>(2.29)   | <b>-0.79</b><br>(-3.82) |
| Inverse Price                          | <b>0</b><br>(16.01)     | <b>0</b><br>(3.39)      | <b>0</b><br>(3.15)      | <b>0.01</b><br>(3.06)   | <b>0.03</b><br>(5.98)   | <b>0.03</b><br>(5.02)   | <b>0.03</b><br>(4.66)   | <b>0.02</b><br>(4.32)   | <b>0.09</b><br>(2.71)   |
| Market Vol                             | <b>0</b><br>(9.07)      | <b>0</b><br>(8.12)      | <b>0</b><br>(5.64)      | <b>0</b><br>(2.9)       | <b>-0.01</b><br>(-6.65) | <b>-0.01</b><br>(-5.68) | <b>-0.01</b><br>(-5.49) | <b>-0.01</b><br>(-5.4)  | <b>0.03</b><br>(2.07)   |
| AT Intensity                           | <b>0</b><br>(-2.47)     | 0<br>(-1.22)            | 0<br>(-1.31)            | 0<br>(-0.3)             | 0<br>(1.2)              | 0<br>(0.46)             | 0<br>(0.57)             | 0<br>(0.64)             | -0.01<br>(-1.08)        |
| Rollover                               | <b>0</b><br>(-1.98)     | 0<br>(-1.42)            | -0.01<br>(-1.41)        | 0.01<br>(0.22)          | 0.01<br>(0.68)          | -0.02<br>(-0.73)        | -0.01<br>(-0.44)        | -0.01<br>(-0.54)        | <b>-0.45</b><br>(-1.98) |
| Excluded                               | 0.01<br>(1.83)          | <b>0.05</b><br>(2.87)   | <b>0.07</b><br>(2.68)   | <b>0.16</b><br>(2.25)   | <b>-0.37</b><br>(-2.63) | <b>-0.39</b><br>(-2.5)  | <b>-0.42</b><br>(-2.63) | <b>-0.44</b><br>(-2.78) | <b>1.7</b><br>(2.17)    |
| Adjusted R <sup>2</sup>                | 0.67                    | 0.34                    | 0.33                    | 0.12                    | 0.45                    | 0.35                    | 0.33                    | 0.31                    | 0.13                    |
| Treated                                | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      |
| Control                                | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      | 41                      |
| # of obs.                              | 9515                    | 9512                    | 9435                    | 8304                    | 9515                    | 9515                    | 9515                    | 9515                    | 9515                    |

**Table 9** Difference-in-Difference estimates for impact on efficiency, Event 1

The table reports the results of DiD estimations results for on efficiency variables:  $\sigma_{IC,250k}$ ,  $\sigma_{IC,500k}$ ,  $\sigma_{IC,1000k}$  and  $|VR - 1|$  along with  $\sigma_r$  as a measure of volatility.

Panel A presents the results for matched treated stocks on SSF market relative to the matched control stocks on the spot market. Panel B presents the results for matched treated and controls sets on the spot market.

‘Fee’ is the dummy variable differentiating treated and control, and ‘Treated’ differentiates between the pre-OTR fee and the post-OTR fee periods. ‘Treated  $\times$  Fee’ is the interaction term that captures the causal effect of the fee on the OTR for the treated sample.

The  $t$ -statistics based on standard errors clustered by stock and time are presented in parentheses. **Boldface** values indicate significance at 5%.

|                                        | $\sigma_r$              | $\sigma_{IC,250k}$      | $\sigma_{IC,500k}$      | $\sigma_{IC,1000k}$    | $ VR - 1 $               |
|----------------------------------------|-------------------------|-------------------------|-------------------------|------------------------|--------------------------|
| Panel A: SSF (treated) - Spot(control) |                         |                         |                         |                        |                          |
| Fee                                    | 0.41<br>(0.81)          | 0<br>(-0.44)            | 0<br>(0.44)             | 0.02<br>(1.66)         | 0<br>(-0.81)             |
| Treated                                | <b>12.59</b><br>(5.95)  | -0.01<br>(-0.36)        | -0.02<br>(-0.77)        | <b>0.07</b><br>(3.42)  | <b>-0.16</b><br>(-19.84) |
| Treated $\times$ Fee                   | <b>-7.47</b><br>(-5.73) | <b>-0.05</b><br>(-4.15) | <b>-0.06</b><br>(-4.55) | <b>-0.11</b><br>(-6.9) | <b>0.01</b><br>(2.28)    |
| Market Cap                             | -2.09<br>(-1.87)        | -0.01<br>(-1.45)        | -0.01<br>(-1.14)        | -0.01<br>(-1.34)       | 0.01<br>(1.14)           |
| Inverse Price                          | -0.07<br>(-0.72)        | <b>0</b><br>(-2.78)     | <b>0</b><br>(-2.97)     | <b>0</b><br>(-2.25)    | <b>0</b><br>(0.89)       |
| Market Vol                             | <b>0.47</b><br>(12.75)  | <b>0</b><br>(8.7)       | <b>0</b><br>(8.15)      | <b>0</b><br>(8.68)     | <b>0</b><br>(-6.88)      |
| AT Intensity                           | -0.09<br>(-1.65)        | 0<br>(-0.25)            | 0<br>(0.4)              | 0<br>(-0.16)           | 0<br>(-0.01)             |
| Rollover                               | 1.1<br>(1.66)           | <b>0.02</b><br>(2.04)   | 0.02<br>(1.44)          | <b>0.04</b><br>(2.61)  | 0<br>(-0.61)             |
| Adjusted R <sup>2</sup>                | 0.27                    | 0.11                    | 0.09                    | 0.08                   | 0.52                     |
| Treated                                | 39                      | 39                      | 39                      | 39                     | 39                       |
| Control                                | 39                      | 39                      | 39                      | 39                     | 39                       |
| # of obs.                              | 6060                    | 6058                    | 6034                    | 5720                   | 6060                     |
| Panel B: Spot(treated)-Spot(control)   |                         |                         |                         |                        |                          |
| Fee                                    | -0.76<br>(-1.56)        | 0<br>(-0.66)            | 0<br>(0.31)             | 0.01<br>(1.36)         | 0<br>(0.48)              |
| Treated                                | <b>-3.3</b><br>(-2.72)  | -0.04<br>(-1.9)         | <b>-0.05</b><br>(-2.22) | 0.01<br>(0.44)         | 0<br>(0.68)              |
| Treated $\times$ Fee                   | 0.45<br>(0.75)          | -0.01<br>(-0.64)        | -0.01<br>(-0.74)        | 0<br>(0.05)            | 0<br>(-0.5)              |
| Market Cap                             | -0.82<br>(-1.2)         | -0.01<br>(-1.38)        | -0.01<br>(-1.09)        | -0.01<br>(-0.94)       | 0<br>(1.87)              |
| Inverse Price                          | -0.02<br>(-0.26)        | <b>0</b><br>(-2.7)      | <b>0</b><br>(-3.56)     | <b>0</b><br>(-4.06)    | <b>0</b><br>(8.45)       |
| Market Vol                             | <b>0.27</b><br>(15.07)  | <b>0</b><br>(8.32)      | <b>0</b><br>(8.07)      | <b>0</b><br>(8.83)     | <b>0</b><br>(-7.01)      |
| AT Intensity                           | <b>-0.08</b><br>(-2.92) | 0<br>(-1.14)            | 0<br>(-0.64)            | <b>0</b><br>(-2.01)    | 0<br>(-0.21)             |
| Rollover                               | -0.42<br>(-1.96)        | <b>-0.01</b><br>(-2.9)  | <b>-0.01</b><br>(-2.22) | 0<br>(-0.44)           | 0<br>(1.65)              |
| Adjusted R <sup>2</sup>                | 0.16                    | 0.1                     | 0.09                    | 0.04                   | 0.1                      |
| Treated                                | 39                      | 39                      | 39                      | 39                     | 39                       |
| Control                                | 39                      | 39                      | 39                      | 39                     | 39                       |
| # of obs                               | 6715                    | 6713                    | 6689                    | 6358                   | 6715                     |



## Event 1

We find that the coefficients of the interaction term, ‘Treated  $\times$  Fee’ are significant and negative in Panel A for all the liquidity risk measures –  $\sigma_{IC,250k}$ ,  $\sigma_{IC,500k}$ ,  $\sigma_{IC,1000k}$ . No such impact is observed in Panel B. This suggests that the liquidity risk improved as a result of the OTR FEE for the SSF but there is no similar change in the underlying spot market relative to the non-SSF stocks. The coefficient on the informational efficiency measure ( $|VR - 1|$ ) is positive and significant in Panel A and insignificant in Panel B. This shows a relative worsening of the efficiency of the treated stocks on the SSF market.

Overall, our results suggest that the fee reduced liquidity volatility for the treated SSF, with no such impact on the underlying stock markets. This is a benefit of the OTR fee in 2009. However, the variance ratio measure shows that the informativeness of SSF prices has decreased.

## Event 2

In comparison to Event 1, we do not see a consistent change in any of the efficiency measures based on the coefficients of the interaction term ‘Treated $\times$ Fee, in Table 10. This is true for both the treated stocks on the SSF market (Panel A) as well as the treated stocks on the spot market (Panel B). Thus, there is no evidence that the OTR FEE imposed by SEBI in 2012 had any effect, either positive or negative, on the efficiency of the market.

## 5.4 Summary

Our results are a significant departure from earlier literature that shows a negative impact of such regulatory interventions on the overall market quality (Friederich and Payne, 2015; Malinova *et al.*, 2013). Our findings suggest that the OTR FEE of Event 1 improved the overall market quality in terms of liquidity as well as efficiency measures. However, there was no significant impact of this fee after Event 2 on the OTR levels or market efficiency. We do see, however, that the liquidity of the market has improved for transactions of large trade sizes of a million rupees. Both these episodes of OTR fee imposition suggest that the intention of the intervention shapes the exact form and design of how the OTR fee is applied which, in turn, shapes the impact of the fee on the behaviour of algorithmic traders, as well as on resultant market quality.

## 6 Conclusion

Financial market regulators world wide have mandated the use of charges and fees as a mechanism to manage the perception of excessive trading in these markets. These interven-

**Table 10** Difference-in-Difference estimates for impact on efficiency, Event 2

The table reports the results of DiD estimations results on efficiency variables:  $\sigma_{IC,250k}$ ,  $\sigma_{IC,500k}$ ,  $\sigma_{IC,1000k}$  and  $|VR - 1|$  along with  $\sigma_r$  as a measure of volatility.

Panel A presents the results for treated stocks on the SSF market relative the matched control stocks on the spot market. Panel B presents the results for treated and matched controls sets on the spot market.

‘Fee’ is the dummy variable differentiating treated and control, and ‘Treated’ differentiates between the pre-OTR fee and post-OTR fee periods. ‘Treated  $\times$  Fee’ is the interaction term that captures the causal effect of the fee on the OTR for the treated sample.

The  $t$ -statistics based on standard errors clustered by stock and time are provided in parentheses. **Boldface** values indicate significance at 5%.

|                                        | $\sigma_r$   | $\sigma_{IC,250k}$ | $\sigma_{IC,500k}$ | $\sigma_{IC,1000k}$ | $ VR - 1 $   |
|----------------------------------------|--------------|--------------------|--------------------|---------------------|--------------|
| Panel A: SSF (treated) - Spot(control) |              |                    |                    |                     |              |
| Fee                                    | <b>-1.41</b> | <b>-0.03</b>       | <b>-0.03</b>       | 0                   | 0            |
|                                        | (-3.07)      | (-3.21)            | (-2.39)            | (-0.18)             | (-0.2)       |
| Treated                                | <b>12.58</b> | -0.02              | -0.03              | <b>0.07</b>         | <b>-0.13</b> |
|                                        | (6.52)       | (-1.28)            | (-1.19)            | (4.21)              | (-17.69)     |
| Treated $\times$ Fee                   | <b>-5.57</b> | 0                  | -0.01              | <b>-0.05</b>        | 0.01         |
|                                        | (-2.99)      | (-0.07)            | (-0.59)            | (-2.96)             | (1.36)       |
| Market Cap                             | -1.15        | -0.01              | -0.01              | 0                   | <b>0.01</b>  |
|                                        | (-1.78)      | (-1.33)            | (-0.8)             | (-0.24)             | (3.2)        |
| Inverse Price                          | <b>0.26</b>  | 0                  | 0                  | <b>0</b>            | <b>0</b>     |
|                                        | (7.11)       | (1.18)             | (1.25)             | (2.49)              | (9.77)       |
| Market Vol                             | <b>0.09</b>  | <b>0</b>           | <b>0</b>           | <b>0</b>            | 0            |
|                                        | (4.31)       | (3.63)             | (2.02)             | (3.29)              | (-0.22)      |
| AT Intensity                           | <b>-0.09</b> | 0                  | 0                  | 0                   | 0            |
|                                        | (-3.49)      | (-0.86)            | (0.45)             | (-1.42)             | (1.52)       |
| Rollover                               | -0.41        | 0.01               | -0.01              | 0.01                | -0.01        |
|                                        | (-0.92)      | (1.34)             | (-1.12)            | (0.7)               | (-1.36)      |
| Excluded                               | <b>7.71</b>  | <b>0.04</b>        | <b>0.06</b>        | <b>0.08</b>         | -0.01        |
|                                        | (3.03)       | (2.47)             | (3.08)             | (3.52)              | (-0.94)      |
| Adjusted R <sup>2</sup>                | 0.45         | 0.08               | 0.03               | 0.13                | 0.38         |
| Treated                                | 41           | 41                 | 41                 | 41                  | 41           |
| Control                                | 41           | 41                 | 41                 | 41                  | 41           |
| # of obs                               | 7485         | 7482               | 7388               | 6393                | 7485         |
| Panel B: Spot(treated)-Spot(control)   |              |                    |                    |                     |              |
| Fee                                    | <b>-1.16</b> | <b>-0.02</b>       | <b>-0.03</b>       | 0                   | 0            |
|                                        | (-2.73)      | (-2.98)            | (-2.18)            | (0.57)              | (0.04)       |
| Treated                                | -1.43        | -0.02              | -0.01              | <b>0.09</b>         | 0            |
|                                        | (-1.78)      | (-1.2)             | (-0.61)            | (3.26)              | (0.98)       |
| Treated $\times$ Fee                   | -0.65        | 0                  | -0.01              | -0.05               | 0.01         |
|                                        | (-1.07)      | (-0.44)            | (-0.79)            | (-1.83)             | (1.72)       |
| Market Cap                             | <b>-0.66</b> | -0.01              | 0                  | 0                   | <b>0.01</b>  |
|                                        | (-2.13)      | (-1.25)            | (-0.81)            | (-0.38)             | (3.19)       |
| Inverse Price                          | <b>0.29</b>  | 0                  | 0                  | 0                   | <b>0</b>     |
|                                        | (11.62)      | (1.57)             | (0.92)             | (0.04)              | (13.35)      |
| Market Vol                             | <b>0.11</b>  | <b>0</b>           | <b>0</b>           | <b>0</b>            | 0            |
|                                        | (7.69)       | (5.72)             | (2.9)              | (3.03)              | (-1.2)       |
| AT Intensity                           | -0.02        | 0                  | 0                  | 0                   | <b>0</b>     |
|                                        | (-1.89)      | (-0.37)            | (0.39)             | (-0.88)             | (4.98)       |
| Rollover                               | -0.33        | 0                  | 0                  | 0.01                | <b>-0.01</b> |
|                                        | (-1.55)      | (0.5)              | (-0.57)            | (1.02)              | (-2.96)      |
| Excluded                               | <b>2.27</b>  | <b>0.05</b>        | <b>0.08</b>        | 0.08                | <b>-0.02</b> |
|                                        | (2.47)       | (3.36)             | (3.32)             | (1.67)              | (-2.96)      |
| Adjusted R <sup>2</sup>                | 0.58         | 0.09               | 0.03               | 0.04                | 0.21         |
| Treated                                | 41           | 41                 | 41                 | 41                  | 41           |
| Control                                | 41           | 41                 | 41                 | 41                  | 41           |
| # of obs                               | 9515         | 9512               | 9415               | 8223                | 9515         |

tions have been increasingly used, the more that trading on financial markets are driven by algorithms. We use a unique opportunity of two events to study how effective such regulatory interventions are in causing changes in order placement and trading choices, as well as in causing changes in overall market quality. In these two events, the OTR fee was used to control inadvertent, adverse effects of algorithmic trading in the Indian equity market as such trading became a higher fraction of overall trading activity. The first event was when the exchange used the fee to manage bandwidth load, and the second came later when the regulator imposed the fee in response to public policy concerns.

This opportunity is unique because the two events play out in the same micro-structure, but are clearly separated in time. Further, the markets observed are highly liquid, so that market quality can be measured in a statistically robust manner. Finally, there are market micro-structure elements that allow us to frame the difference-in-difference regressions in an innovative set of treated and matched control samples to strengthen the inference of the causal impact of the fee.

Our analysis shows that the exchange imposed OTR fee, which was applied with an explicit objective, was effective while there is ambiguity in impact of the fee imposed by the regulator. The aggregate market OTR decreased, on average, when the exchange imposed the fee, while a similar, simple examination suggests that the same measure of OTR *increased* on average when the regulator imposed the fee. This is counter to the apparent objective of the intervention. We performed a more detailed empirical analysis that shows a differential impact of the fee at different parts of the limit order book. However, such an asymmetric impact defeats the purpose of the intervention, which has to be simple and has to deliver readily visible results to achieve the objective of alleviating public policy concerns.

These results show that it is not just the intervention, but rather the clarity of the objective for which it is applied and, therefore, the specifics of the design of the intervention that is important in delivering desired outcomes. In the case of the exchange intervention that was used to disincentivise spurious order flow, there was an overall reduction of the OTR *and* an improvement in the liquidity and efficiency in the derivatives markets that it targeted. Both the estimated impact cost of a trade and the expected volatility of impact cost are reduced. On the other hand, there is ambiguity in the impact of the fee imposed by the regulator.

If the objective of the regulator in imposing the OTR fee is in the public interest, then the evidence of the impact on the OTR or market quality is not likely to boost investor confidence in the market. An intervention in the public interest should benefit from visible results using simple, default analyses. We present a cautionary tale of regulators intervening in market design: optimal outcomes are best guided with clear and focused objectives. Our analysis suggests that the regulators must state what the expected outcomes are as part of the objective and design of a market intervention, and that these outcomes should be readily measurable so as to be easily visible to the public.

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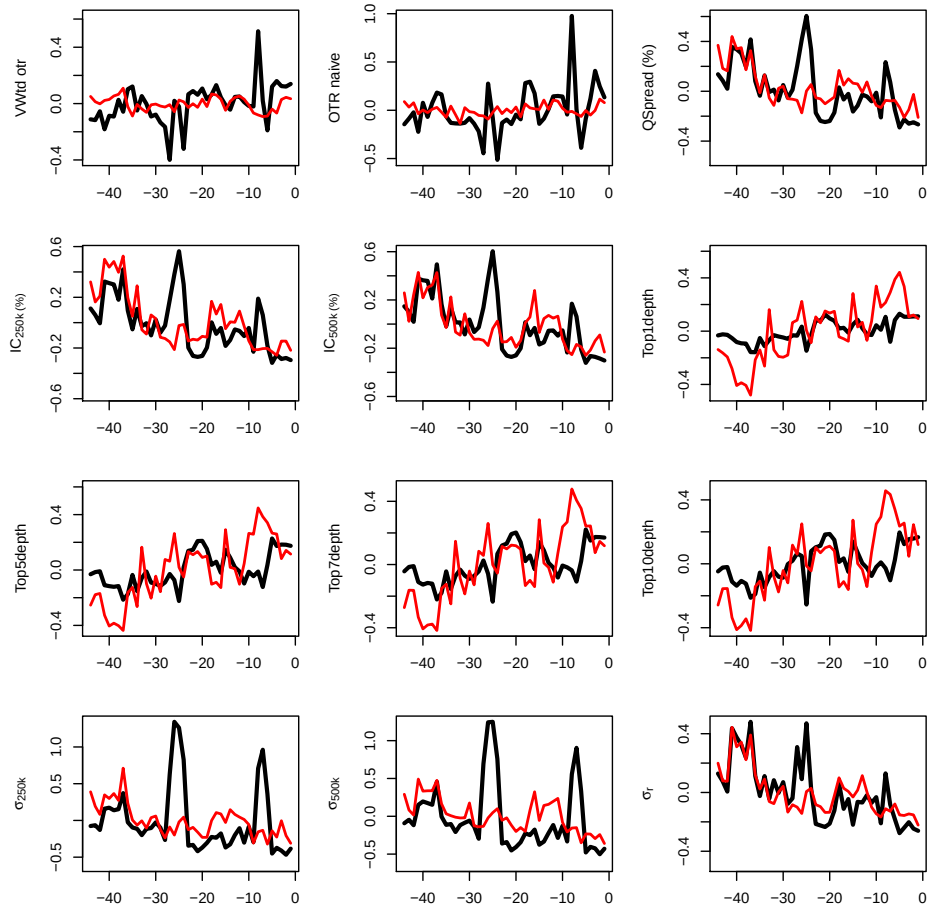
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**Figure 7** Pre-treatment outcome variables for matched treated stocks on the futures market versus control stocks the on spot market in Event 1

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The figure shows the evolution of outcome variables prior to the treatment for Event 1. For each variable, we plot the cross-sectional average for treated (black line) and control stocks (red line), minus the respective pre-event average. This approach is similar to that of Colliard and Hoffmann (2017). The graphs are shown for variables on the futures market for the treated set and the spot market as the matched control set.

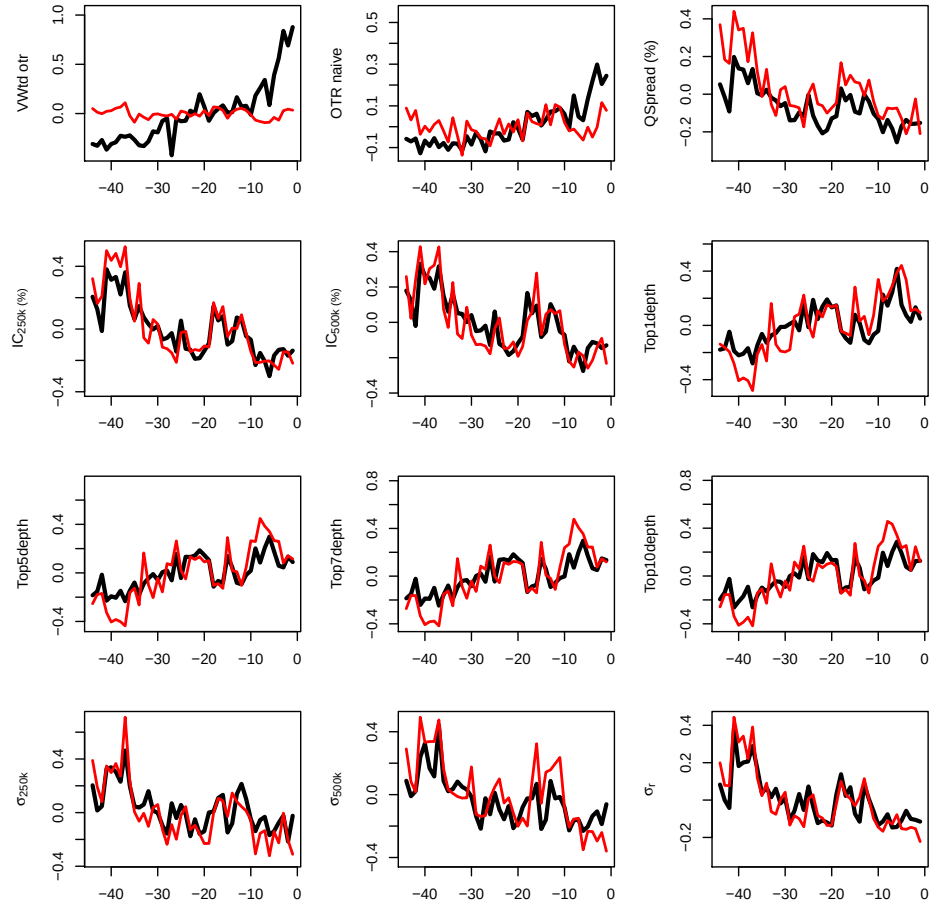


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**Figure 8** Pre-treatment outcome variables for treated stocks with futures on the spot market versus matched controls on the spot market in Event 1

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The figure shows the evolution of outcome variables prior to the treatment for Event 1. For each variable, we plot the cross-sectional average for treated (black line) and control stocks (red line), minus the respective pre-event average. The graphs are shown for variables on the spot market for the treated set and the spot market for the matched control set.

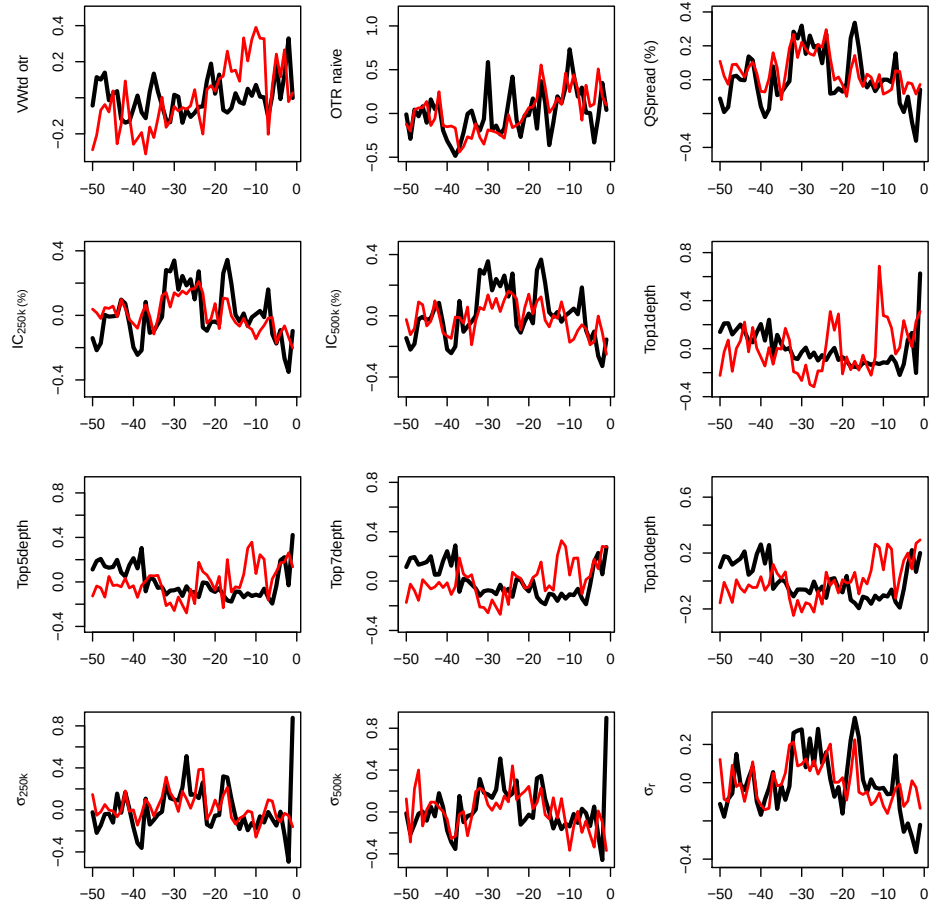


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**Figure 9** Pre-treatment outcome variables for treated stocks on the futures market versus matched control stocks on the spot market in Event 2

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The figure shows the evolution of outcome variables prior to the treatment for Event 2. For each variable, we plot the cross-sectional average for treated (black line) and control stocks (red line), minus the respective pre-event average. This approach is similar to that of Colliard and Hoffmann (2017). The graphs are shown for variables on the futures market for the treated set and the spot market for the matched control set.

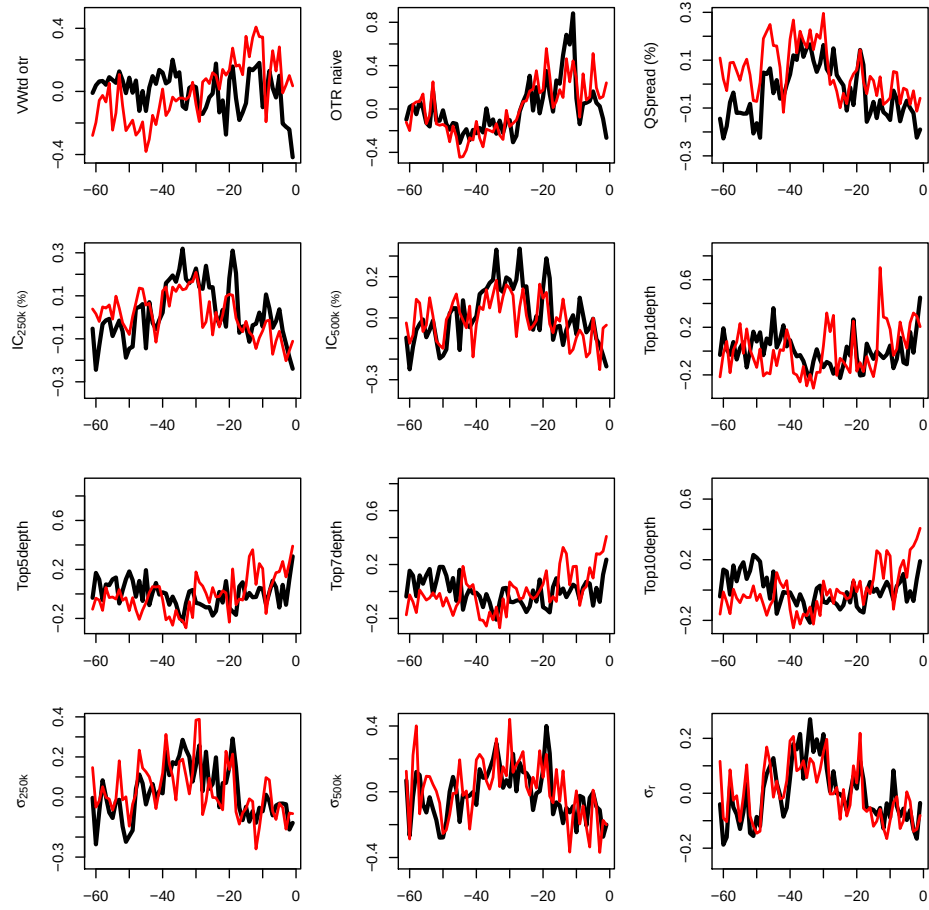


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**Figure 10** Pre-treatment outcome variables for treated stocks with futures on the spot market versus matched control stocks on the spot market in Event 2

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The figure shows the evolution of outcome variables prior to the treatment for Event 2. For each variable, we plot the cross-sectional average for treated (black line) and control stocks (red line), minus the respective pre-event average. The graphs are shown for variables on the spot market for the treated set and the spot market for the matched control set.





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